Online Supplement to Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

This supplement retains the existing proofs and complete principal adverse-result inventories, with links to the unchanged longer records. It accompanies the submission-final manuscript and retains the accepted fixed simulation, same-dealer daily aggregation and fixed-parameter full-prefix backcast. Each addition was fixed before its own new output within a post-result / exploratory revision. The two original primary comparisons retain their values and roles. The earlier calibrated design and stability screen retain separate sections. The v10 additions report paired menu isolation, independently reported CFTC categories, complete model confidence sets and fixed limit-cost regions.

Tables format saved native fields. Loss percentages are the existing relative differences against the named reference; probabilities are labeled as such. Different source-reported monetary units are not pooled. NA and failed rows remain distinct from complete results.

The complete earlier supplement, full comparison families, all candidate tables, annual and sparse sensitivities, numerical details and source guides remain accessible through EXTENDED RESULTS. This index is an access route, not a substitute for compatible input data or independent validation.

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S1 Known-floor projection and strict quantile coverage

We forecast the sampled maximum of the absolute position, including the observed issue position. Write $B = |N_t|$ and let $Y \geq B$ almost surely, with B measurable at the forecast origin. For $0 < \tau < 1$, the pinball score is $\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+$. A lower inverse conditional quantile is $q_\tau = \inf\{c : F(c \mid \mathcal{F}_t) \geq \tau\}$.

Proposition 1: floor projection. Replacing a report c by $\max(B, c)$ weakly reduces its score at every possible outcome. At the population conditional quantile, the strict breach probability is at most $1-\tau$. These are distinct statements: projection applies to any report; the tail qualification requires the true conditional quantile. Here $q = q_\tau$, and α denotes the stated nominal tail bound.

$$\rho_\tau(Y - c) - \rho_\tau(Y - \max\{B, c\}) = \tau(B - c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \leq \alpha.$$

Proof. If $c \geq B$, the replacement changes nothing. If $c < B$, both errors $Y - c$ and $Y - B$ are nonnegative, so their score difference is $\tau(B - c)$. This proves the pointwise identity without a density assumption. Right continuity of the conditional distribution gives $F(q_\tau \mid \mathcal{F}_t) \geq \tau$, hence $\Pr(Y > q_\tau \mid \mathcal{F}_t) \leq 1 - \tau$. An atom at the quantile can make this strict; exact nominal hit frequency is not required for a valid upper quantile. A fitted report does not automatically inherit this population claim. *QED*

Why the floor matters. The sampled maximum equals the current magnitude whenever the recorded future path remains inside its current absolute-position band. This can create an atom even when later signed observations are continuous. The existence of a floor alone does not imply an atom: an outcome uniformly distributed above the floor provides a counterexample. Equality-based coverage tests therefore diagnose their stated equalities, rather than every possible valid upper-quantile report.

Endpoint marginals do not identify the peak. The exact artificial constructions in the formal example record hold each future-time marginal fixed while changing temporal dependence. Comonotone and disjoint rare events then give different peak quantiles. The example is a logical construction, not another financial dataset or an estimated mechanism. Forecast evaluation must use the ordered path response and the information available at its actual origin.

Takeaway. Known-floor projection is an elementary pointwise improvement. It does not establish empirical calibration, identify the conditional distribution, or turn a fitted endpoint forecast into a forecast of a path maximum.

S2 Classical endpoint-to-peak bound

Proposition 2: the classical endpoint bound. Let future real increments $X_1, \ldots, X_H$ be independent and individually symmetric about zero. Set $S_0 = 0$, $S_h = \sum_{j=1}^{h} X_j$, and fix a starting position x, with absolute magnitude B. Equal increment laws and finite moments are unnecessary. For the population absolute-displacement quantile at level $1 - \alpha/2$, $q = \inf\{c : \Pr(|S_H| \le c) \ge 1 - \alpha/2\}$, the classical scalar maximal inequality and the triangle inequality give

$$\Pr\left(\max_{0\le h\le H} |x + S_h| > B + q\right) \le \Pr\left(\max_{h} |S_h| > q\right) \le 2\Pr(|S_H| > q) \le \alpha.$$

This is a direct application of the classical Lévy inequality, not a new maximal inequality.

Proof, including strict events and atoms. For $c \ge 0$, let η be the first index with $|S_\eta| > c$. On $\{\eta = k, S_k > c\}$, the remaining sum $R_k = S_H - S_k$ is independent of the information through k and is symmetric. Therefore $\Pr(R_k \ge 0) \ge 1/2$, including a possible atom at zero, and that weak sign keeps $S_H > c$. On the corresponding negative crossing, $R_k \le 0$ keeps $S_H < -c$, also with probability at least one half. Summing over the disjoint crossing events gives $\Pr(\max_h |S_h| > c) \le 2\Pr(|S_H| > c)$. The argument covers $c = 0$. Finally, $\max_{0\le h\le H} |x + S_h| \le |x| + \max_h |S_h|$, and right continuity gives the required strict tail bound at q. *QED*

Conditional use and empirical implementation. For an observed random anchor, the same argument must hold under its forecast-origin conditional law, with the corresponding conditional population endpoint quantile. Conditional independence and symmetry about zero cannot be inferred from unconditional marginals. The empirical endpoint plug-in adds the current floor to a historical absolute-displacement order statistic. Overlapping labels, estimated quantiles and changing dependent histories provide no automatic inheritance of this population bound or a pinball-risk ordering.

Two different failure mechanisms. Dependence can destroy the inequality: take $X_1 = Z$, $X_2 = -Z$, with $Z \in \{-1, 1\}$ equally likely. The endpoint is zero, while the sampled maximum is one. Separately, replacing the required absolute-displacement quantile by a generic one-sided endpoint quantile can fail even for independent symmetric increments. The exact finite example is preserved in the formal example record. These logical counterexamples are separate from the Gaussian Monte Carlo cases below; a nonviolating simulation does not prove a theorem extension.

Takeaway. The bound supplies a clear population benchmark under explicit assumptions; the empirical report remains a model to validate, not a coverage certificate.

S3 Monte Carlo mechanisms: original increment persistence

These are the existing fixed-seed Gaussian experiments, not newly run simulations or calibrated replicas of either financial source. Every reported rate is a probability, not a percentage. The peak includes the initial level; the endpoint event concerns absolute displacement from that initial level. Each path cell has 100000 paths. L/Stat starts the level AR in stationarity; L/Abs fixes its absolute-level tail anchor; L/Sig is the separate signed-level tail-anchor sensitivity. I/Zero accumulates stationary AR increments from a zero position. M denotes a stationary-marginal endpoint threshold; C denotes a distinct fixed-start conditional endpoint oracle. Threshold partners sharing a seed use the same paths, not independent repetitions. Tables S1–S3 retain the original increment, corrected primary and secondary simulation results.

Table S1: Original increment-autoregression simulation, unchanged. Serial persistence here concerns increments, not fitted position levels.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.00	I/Zero	20267023	5.490	3035	0.03035	0.00054	0.02454
6	0.70	I/Zero	20267093	10.092	2597	0.02597	0.00050	0.02573
6	0.85	I/Zero	20267108	11.634	2550	0.02550	0.00050	0.02550
6	0.98	I/Zero	20267121	13.190	2557	0.02557	0.00050	0.02557
24	0.00	I/Zero	20285023	10.981	3819	0.03819	0.00061	0.02504
24	0.70	I/Zero	20285093	24.599	2772	0.02772	0.00052	0.02475
24	0.85	I/Zero	20285108	33.394	2619	0.02619	0.00051	0.02525
24	0.98	I/Zero	20285121	49.784	2572	0.02572	0.00050	0.02572

The independent-increment rows satisfy the classical premise. Positive increment coefficients violate independence without necessarily causing a numerical breach of the bound. Their favorable rates are not a general result for serial dependence.

S4 Mean-reverting levels and negative-increment sensitivities

The corrected level process is $N_h = \phi N_{h-1} + \varepsilon_h$, with independent Gaussian innovations of variance $1 - \phi^2$ and unit stationary level variance. Its stationary increments have lag correlation $\rho_X(k) = -(1 - \phi)\phi^{k-1}/2$, so this mechanism differs from a positively persistent increment AR. The fitted financial coefficient describes levels, making the level equation the relevant illustrative mechanism. The negative-increment family is a separate sensitivity, with stationary increments and nonstationary cumulative levels.

Table S2: All corrected primary cases. Mean-reverting level rates are conservative in these finite cases; the negative-increment result must also be retained.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.68	L/Stat/M	20567123	3.009	175	0.00175	0.00013	0.02462
6	0.68	L/Abs/M	20567124	3.009	0	0.00000	0.00000	0.10447
6	0.85	L/Stat/M	20567133	2.502	300	0.00300	0.00017	0.02553
6	0.85	L/Abs/M	20567134	2.502	14	0.00014	0.00004	0.08227
6	0.98	L/Stat/M	20567143	1.071	913	0.00913	0.00030	0.02434
6	0.98	L/Abs/M	20567144	1.071	372	0.00372	0.00019	0.03623
6	-0.30	I/Zero/M	20567223	4.244	3563	0.03563	0.00059	0.02524
6	-0.60	I/Zero/M	20567233	3.127	4439	0.04439	0.00065	0.02558
24	0.68	L/Stat/M	20585123	3.170	636	0.00636	0.00025	0.02495
24	0.68	L/Abs/M	20585124	3.170	1	0.00001	0.00001	0.11416
24	0.85	L/Stat/M	20585133	3.138	402	0.00402	0.00020	0.02520
24	0.85	L/Abs/M	20585134	3.138	1	0.00001	0.00001	0.11159
24	0.98	L/Stat/M	20585143	1.965	768	0.00768	0.00028	0.02468
24	0.98	L/Abs/M	20585144	1.965	85	0.00085	0.00009	0.06201
24	-0.30	I/Zero/M	20585223	8.167	4281	0.04281	0.00064	0.02476
24	-0.60	I/Zero/M	20585233	5.701	5032	0.05032	0.00069	0.02516

All saved level cases have peak rates below the nominal threshold. The largest negative-increment point estimate occurs at the longer horizon and the more negative coefficient. Its rate is 0.05032, with a saved excess of 0.032 percentage points and Monte Carlo SE 0.00069. This does not demonstrate a

population failure. General dependence invalidates the classical premise regardless of the observed direction. Zero observed counts also do not establish zero population probability.

S5 All secondary simulation sensitivities

For stationary random starts, absolute endpoint displacement has variance $2(1-\phi^H)$. At a fixed start x, its conditional Gaussian mean and variance are $(\phi^H-1)x$ and $1-\phi^{2H}$. Thus holding the stationary-marginal endpoint threshold across starts does not condition on a large anchor. The conditional-oracle sensitivities below are distinct thresholds on shared paths; neither is an oracle for the maximum.

Table S3: All secondary anchor and conditional-threshold sensitivities. Conditional and marginal endpoint calibration remain distinct.

Horizon	Persistence ϕ	Process	Seed	Endpoint threshold	Peak breaches	Peak breach probability	Monte Carlo standard error	Endpoint breach probability
6	0.68	L/Abs/C	20567124	3.717	0	0.00000	0.00000	0.02411
6	0.68	L/Sig/M	20567125	3.009	2	0.00002	0.00001	0.06261
6	0.68	L/Sig/C	20567125	3.433	1	0.00001	0.00001	0.02476
6	0.85	L/Abs/C	20567134	3.036	0	0.00000	0.00000	0.02488
6	0.85	L/Sig/M	20567135	2.502	21	0.00021	0.00005	0.05689
6	0.85	L/Sig/C	20567135	2.840	4	0.00004	0.00002	0.02607
6	0.98	L/Abs/C	20567144	1.146	215	0.00215	0.00015	0.02503
6	0.98	L/Sig/M	20567145	1.071	464	0.00464	0.00021	0.03184
6	0.98	L/Sig/C	20567145	1.117	340	0.00340	0.00018	0.02521
24	0.68	L/Abs/C	20585124	3.920	0	0.00000	0.00000	0.02567
24	0.68	L/Sig/M	20585125	3.170	9	0.00009	0.00003	0.06486
24	0.68	L/Sig/C	20585125	3.605	1	0.00001	0.00001	0.02514
24	0.85	L/Abs/C	20585134	3.880	0	0.00000	0.00000	0.02495
24	0.85	L/Sig/M	20585135	3.138	3	0.00003	0.00002	0.06306
24	0.85	L/Sig/C	20585135	3.571	0	0.00000	0.00000	0.02484
24	0.98	L/Abs/C	20585144	2.298	20	0.00020	0.00004	0.02518
24	0.98	L/Sig/M	20585145	1.965	146	0.00146	0.00012	0.04607
24	0.98	L/Sig/C	20585145	2.179	56	0.00056	0.00007	0.02556

The complete correction contains 34 threshold rows on 22 initialized path samples. The retained CSV and portable data record provide exact endpoint moments, deviations from the nominal probability, and the single-prespecified-row bound for zero counts. None estimates financial innovation laws, changing fitted parameters or empirical-quantile uncertainty.

S6 Empirical endpoint plug-in: all original cells

The endpoint plug-in uses each completed historical episode's absolute endpoint displacement and adds its specified empirical quantile to the actual current-position floor. It shares the historical rule's completed origin pool and fixed refit schedule, with no fitted tuning parameter. The operational empirical rank and overlapping financial history are distinct from the known population quantile in the classical proposition. The report therefore has to earn its financial coverage and score performance empirically. Tables S4–S5 report every endpoint plug-in comparison and the separate working-reference contrasts.

Table S4: Every original-cell endpoint plug-in comparison. A population bound does not imply a lower empirical pinball score.

Source	Horizon / history	Endpoint vs historical (%)	Endpoint vs regression (%)	Breaches endpoint / historical
FX	6/720	30.93	33.76	101/332
FX	6/2880	30.39	52.98	75/360
FX	24/720	10.85	-15.71	136/427
FX	24/2880	7.94	12.16	174/357
Bills	6/52	3.37	-43.62	10/11
Bills	6/260	-11.77	-41.49	15/28
Bills	24/52	26.32	-61.42	42/33
Bills	24/260	9.87	-47.64	22/12

Table S5: All separate endpoint-plug-in working-reference contrasts. Adverse score comparisons remain alongside lower-breach reports.

Source	Horizon / history	Reference	Loss-test statistic	Normal p-value	Small-sample p-value	Holm-adjusted p-value
FX	6/720	F	7.891	2.988e-15	3.658e-15	4.483e-14
FX	6/720	QR	6.608	3.908e-11	4.37e-11	5.081e-10
FX	6/2880	F	7.790	6.695e-15	8.123e-15	9.373e-14
FX	6/2880	QR	12.832	1.089e-37	3.451e-37	1.743e-36
FX	24/720	F	1.848	0.06462	0.06562	0.5169
FX	24/720	QR	-1.673	0.09434	0.09556	0.6604
FX	24/2880	F	2.058	0.03958	0.04032	0.3958
FX	24/2880	QR	1.530	0.1259	0.1273	0.7002
Bills	6/52	F	0.971	0.3317	0.3422	0.7002
Bills	6/52	QR	-2.778	0.005471	0.006887	0.06018
Bills	6/260	F	-1.345	0.1785	0.1884	0.7002
Bills	6/260	QR	-3.031	0.002441	0.003236	0.02929
Bills	24/52	F	1.569	0.1167	0.1522	0.7002
Bills	24/52	QR	-1.991	0.04645	0.06948	0.418
Bills	24/260	F	1.092	0.2746	0.3183	0.7002
Bills	24/260	QR	-1.516	0.1294	0.1664	0.7002

The family and its doubled-bandwidth sensitivities are separate from the original regression family. The complete tables keep every adverse direction; no negative comparison is removed by describing the population inequality as conservative. The linked records preserve source-reported mean error amounts and every numerical status.

S7 AR residual replay and its adverse comparisons

At each original refit, estimate an intercept-bearing level recursion using each unique mature adjacent transition once. Replaying a historical ordered residual block from the current position gives the exact algebraic identity

$$M_{i,h}(x) = N_{i+h} + \widehat{\phi}^{h}(x - N_i), \qquad M_{i,0}(x) = x.$$

The empirical peak uses the same completed scenario origins, response rank, floor and refit schedule as its matched historical controls. Finite negative or above-unit coefficients were eligible, not clipped; all observed fitted coefficients in the saved run happened to lie inside the unit interval. Their descrip-

tive means are not identified data-generating parameters. Table S6 retains every autoregressive replay comparison, including its adverse breach counts.

Table S6: Complete original autoregressive replay comparison. Lower scores can accompany more strict breaches.

Source	Horizon	History	Mean fitted $\widehat{\phi}$	Replay vs translated (%)	Replay vs regression (%)	Breaches replay / translated
FX	6	720	0.981	-1.54	2.45	282/255
FX	6	2880	0.985	-0.55	16.20	338/268
FX	24	720	0.981	2.39	-25.32	401/289
FX	24	2880	0.985	5.15	8.68	444/292
Bills	6	52	0.676	-13.44	-52.66	22/11
Bills	6	260	0.847	-18.89	-47.21	31/28
Bills	24	52	0.711	35.13	-59.43	56/33
Bills	24	260	0.850	-24.12	-63.86	59/12

The replay improves on original regression in every original Treasury cell but loses in most FX cells. Its comparisons with the translated-path rule have mixed directions, while strict breaches increase in every row above. Sparse-grid and annual signs can reverse. The full paired, annual and fixed-grid tables remain in the AR records. These are post-result / exploratory same-data calculations, not independent validation.

S8 Ordinary quantile ranks and minimizer endpoints

The ordinary empirical quantile and the corrected empirical rule use the same completed pool, refit times and issue floor. The former takes the lower empirical pinball minimizer; the latter uses the specified add-one rank. When the product of the probability level and pool size is an integer, adjacent reports can be the two endpoints of the same empirical minimizer interval. Calling the lower report a uniquely optimal comparator would therefore be wrong. At the short weekly pool, the noninteger rank distinction differs. Table S7 reports the complete rank ablation and its breach counts.

Table S7: Complete ordinary-rank ablation. Lower empirical reports increase breach occasions and need not improve score.

Source	Horizon	History	Regression vs empirical (%)	Empirical vs historical (%)	Breaches empirical / historical / regression
FX	6	720	-1.967	-0.153	337/332/531
FX	6	2880	-14.773	0.008	365/360/376
FX	24	720	31.508	-0.003	432/427/924
FX	24	2880	-3.783	0.026	359/357/531
Bills	6	52	87.086	-1.995	14/11/84
Bills	6	260	46.058	3.237	32/28/54
Bills	24	52	208.258	6.213	37/33/132
Bills	24	260	105.020	2.353	15/12/84

The reconstructed historical summaries match the original controls. The lower report increases the number of strict hits and the mean underprediction amount in every cell, but its loss direction is mixed. Every Treasury regression remains worse than the lower empirical report. This ablation isolates a rank convention on the same inspected record; it neither supplies a fresh validation sample nor establishes

universal superiority of either finite-sample quantile endpoint. Exact counts, annual splits and fixed-grid rows are linked in the complete rank records.

S9 Fixed-report error preferences

For a fixed report, changing the loss weight traces a line $R_a = aU+(1-a)V$. Here, $U = \mathrm{mean}(P-C)_+$ is mean underprediction and $V = \mathrm{mean}(C-P)_+$ is mean overprediction. These are passive error amounts over all origins, not observed funding costs. Breach frequency averages an indicator and need not rank reports like either amount. The following source-reported monetary values are never pooled across FX and weekly aggregates.

Table S8: The four fixed headline comparisons. The longer-horizon FX result is sensitive to error weights.

Comparison	Under-prediction left	Under-prediction right	Mean overprediction left	Mean overprediction right	Equal-loss weight a
fx_h6_primary	156123.03	166531.06	4750560.18	5671390.59	none
fx_h6_matched	156123.03	185672.23	4750560.18	5526087.27	none
fx_h24_legacy	317678.12	149813.88	8658017.18	12121206.97	$\frac{86579744869851975}{90776350660766843}$
treasury_h6_primary	3131.86	1395.64	23688.09	28656.17	$\frac{24840389716843550}{33521504184965797}$

The equality column gives exact saved rational boundaries for the mixture weight defined above, not newly estimated costs. The short-horizon FX primary reduces both mean error amounts; its elementary-score curves can nevertheless cross because a Murphy diagram varies the elementary score threshold, a different comparison from reweighting these two unconditional amounts. The longer-horizon FX legacy pair reverses near the loss-weight ratio reported in the main text. Actual decision costs were not observed.

The linked complete frontiers include every open region and every boundary tie. The full and sparse samples remain separate, and feasible reports outside the specified menu are not part of these menu-relative envelopes. All pairwise relations, dominated labels and exact arithmetic are additionally preserved in the preference records.

S10 Complete preference-frontier source

The complete full-sample and fixed sparse-grid frontiers retain every boundary tie and open region. Their unchanged tables appear in the v9 Supplement, Tables S9–S10, and the exact rational source fields remain in the linked preference results. This supplement retains the error amounts and reversal above, while allocating the full new-method comparison tables below. No preference result was recomputed or removed from the repository.

S11 Complete weekly panel: definitions and first half

Each cell compares the same peak target, completed-label pool, origin sequence, fixed refit schedule and native monetary units. The original full regression, smaller predictor regression, expanded-ridge full regression and ordinary empirical report all remain visible. The additional series are related weekly aggregates, not independent individual-dealer replications. Changes in predictors and changes in the ridge menu are distinct interventions. Every listed comparison is post-result / exploratory.

The historical column gives mean pinball loss and strict hits. Other columns give the saved percentage loss difference against that same historical rule and strict hits; negative percentages favor the column method. Original full regression, four-feature regression and expanded-menu full regression are distinct fitted procedures. I is the ordinary empirical-quantile report, with the endpoint qualification in Section S8. Across the saved panel, expanded full-regression median excess loss is 9.22 percent, compared with 92.63 percent for the original menu. Expanded regression still loses in 34 cells. This improvement does not erase the remaining failures or identify a causal feature effect.

Table S9: Complete panel, first half. Additional regularization leaves substantial adverse cases.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-feature regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSMBS-TOT	6/52	921.7 / 21	170.97 / 101	3.24 / 27	5.82 / 28	3.26 / 27
PDPOSMBS-TOT	6/260	945.6 / 26	7.36 / 19	1.17 / 27	7.36 / 19	1.96 / 27
PDPOSMBS-TOT	24/52	1753.7 / 39	215.55 / 145	5.09 / 43	66.24 / 104	3.49 / 42
PDPOSMBS-TOT	24/260	1611.9 / 25	26.79 / 29	1.04 / 26	0.72 / 24	1.14 / 26
PDPOSCS-TOT	6/52	530.3 / 16	52.75 / 48	-5.79 / 23	-4.70 / 23	-6.28 / 23
PDPOSCS-TOT	6/260	399.7 / 6	1.90 / 15	-1.10 / 6	-0.95 / 8	-1.56 / 7
PDPOSCS-TOT	24/52	579.4 / 16	73.78 / 63	1.85 / 24	2.97 / 25	2.12 / 23
PDPOSCS-TOT	24/260	569.9 / 6	27.46 / 3	-3.38 / 6	-3.06 / 5	-4.68 / 6
PDPOSFGS-TOT	6/52	400.7 / 16	107.22 / 77	-4.15 / 24	-4.29 / 20	-4.98 / 18
PDPOSFGS-TOT	6/260	358.3 / 6	-1.08 / 14	4.88 / 9	-6.04 / 6	-1.54 / 6
PDPOSFGS-TOT	24/52	476.2 / 36	128.67 / 64	13.36 / 37	10.67 / 37	4.52 / 38
PDPOSFGS-TOT	24/260	354.0 / 9	30.02 / 21	72.24 / 26	-0.97 / 10	-1.50 / 9
PDPOSTIPS-L2	6/52	287.3 / 26	217.18 / 125	6.74 / 29	2.64 / 28	1.69 / 26
PDPOSTIPS-L2	6/260	269.3 / 24	66.28 / 45	1.78 / 25	66.28 / 45	1.63 / 25
PDPOSTIPS-L2	24/52	641.0 / 48	307.36 / 181	24.88 / 69	152.16 / 146	1.88 / 49
PDPOSTIPS-L2	24/260	532.2 / 47	226.91 / 147	11.41 / 59	226.91 / 147	1.84 / 51
PDPOSTIPS-G11	6/52	42.3 / 23	196.70 / 118	1.71 / 29	6.68 / 41	-1.13 / 26
PDPOSTIPS-G11	6/260	41.8 / 14	90.31 / 38	63.18 / 29	-1.52 / 13	-2.53 / 14
PDPOSTIPS-G11	24/52	80.8 / 61	218.69 / 123	135.59 / 103	64.63 / 78	-0.01 / 63
PDPOSTIPS-G11	24/260	51.2 / 9	98.59 / 33	37.18 / 30	98.59 / 33	-1.09 / 10
PDPOSTIPS-G2	6/52	181.3 / 16	208.61 / 96	8.58 / 21	35.47 / 41	3.33 / 19
PDPOSTIPS-G2	6/260	179.4 / 19	63.47 / 36	10.78 / 24	63.47 / 36	-0.29 / 19
PDPOSTIPS-G2	24/52	377.9 / 31	277.81 / 131	11.77 / 42	40.47 / 60	-3.63 / 33
PDPOSTIPS-G2	24/260	286.0 / 23	109.33 / 66	120.12 / 58	109.33 / 66	0.29 / 23

S12 Complete weekly panel: second half

The second half follows the identical convention. All original and expanded results remain unchanged. Cross-series dependence, aggregation offsets, native-unit differences and overlapping economic positions preclude treating these cells as independent replications. Forecasts are fitted separately; observations and losses are not pooled into a single financial population.

Table S10: Complete panel, second half. The enlarged penalty menu is not uniformly better than historical rules.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-feature regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSTIPS-G6L11	6/52	133.6 / 9	59.30 / 73	-1.65 / 12	-2.52 / 13	-0.79 / 12
PDPOSTIPS-G6L11	6/260	114.9 / 4	-5.72 / 14	-1.11 / 4	-1.17 / 4	-1.00 / 4

Table S10: Continued.

Series	Horizon / history	Historical loss / breaches	Original regression % / hits	Four-feature regression % / hits	Expanded-menu regression % / hits	Empirical % / breaches
PDPOSTIPS-G6L11	24/52	147.0 / 10	76.48 / 92	-0.97 / 16	-2.43 / 17	0.44 / 15
PDPOSTIPS-G6L11	24/260	133.8 / 2	-26.42 / 1	-1.30 / 2	-1.33 / 2	-0.55 / 2
PDPOSGSC-L2	6/52	1232.9 / 15	215.90 / 129	-3.85 / 23	215.90 / 129	-2.12 / 19
PDPOSGSC-L2	6/260	1247.4 / 15	64.40 / 40	9.04 / 20	2.20 / 20	0.61 / 17
PDPOSGSC-L2	24/52	2270.8 / 46	276.55 / 165	182.80 / 96	131.63 / 93	5.02 / 55
PDPOSGSC-L2	24/260	2918.7 / 33	41.60 / 55	-0.19 / 33	0.12 / 32	0.14 / 34
PDPOSGSC-G2L3	6/52	792.4 / 24	121.15 / 93	145.25 / 112	25.51 / 40	3.86 / 28
PDPOSGSC-G2L3	6/260	1004.0 / 28	21.31 / 44	22.62 / 59	21.31 / 44	0.23 / 30
PDPOSGSC-G2L3	24/52	2101.9 / 47	112.95 / 135	129.55 / 151	112.95 / 135	1.78 / 54
PDPOSGSC-G2L3	24/260	2677.0 / 55	11.60 / 64	39.73 / 91	-2.06 / 52	1.93 / 55
PDPOSGSC-G3L6	6/52	764.9 / 12	87.61 / 82	75.44 / 72	10.10 / 29	-4.25 / 15
PDPOSGSC-G3L6	6/260	699.1 / 20	0.79 / 15	1.91 / 21	0.79 / 15	2.27 / 22
PDPOSGSC-G3L6	24/52	977.4 / 38	544.34 / 137	299.12 / 120	19.44 / 47	9.23 / 45
PDPOSGSC-G3L6	24/260	931.0 / 20	15.52 / 37	0.13 / 20	-5.76 / 17	0.56 / 20
PDPOSGSC-G6L7	6/52	542.3 / 16	253.66 / 142	89.00 / 79	48.88 / 63	-1.24 / 24
PDPOSGSC-G6L7	6/260	564.9 / 20	31.61 / 42	71.05 / 49	31.61 / 42	0.43 / 21
PDPOSGSC-G6L7	24/52	993.4 / 43	264.80 / 217	264.77 / 215	264.80 / 217	11.08 / 47
PDPOSGSC-G6L7	24/260	699.0 / 22	146.41 / 77	142.57 / 67	146.41 / 77	0.98 / 23
PDPOSGSC-G7L11	6/52	505.7 / 25	175.62 / 83	15.72 / 34	86.69 / 55	14.85 / 34
PDPOSGSC-G7L11	6/260	506.2 / 25	8.34 / 31	-0.06 / 25	8.34 / 31	0.44 / 26
PDPOSGSC-G7L11	24/52	1653.9 / 36	94.95 / 105	62.40 / 91	-0.51 / 39	2.38 / 36
PDPOSGSC-G7L11	24/260	1002.9 / 32	229.96 / 94	202.78 / 99	229.96 / 94	8.62 / 35

The expanded-menu selection precedes its evaluation, as documented in the E5 records. The original panel's entire candidate/status inventory remains in the original panel records, and the fitted expanded-menu predictions are not substitutes for unavailable methods or independent dealer data.

S13 Complete added-panel endpoint-plug-in comparison

The original added-panel endpoint plug-in was evaluated on the same task grids as the historical and regression reports. Each entry gives the saved loss percentage against the historical rule, followed by strict hits for the endpoint plug-in and historical report. These are all related aggregate tasks, not independent replications or source-pooled monetary losses.

Table S11: Every additional-panel endpoint-plug-in result. Fewer breaches do not guarantee a better score.

Series	Horizon 6 history 52	Horizon 6 history 260	Horizon 24 history 52	Horizon 24 history 260
PDPOSMBS-TOT	7.17; 11/21	7.33; 5/26	-17.87; 27/39	-3.92; 22/25
PDPOSCS-TOT	18.28; 11/16	17.56; 2/6	-5.06; 11/16	19.63; 4/6
PDPOSFGS-TOT	-4.74; 13/16	16.77; 0/6	-3.72; 37/36	-4.06; 12/9
PDPOSTIPS-L2	-1.89; 12/26	-7.42; 19/24	-34.60; 30/48	-4.52; 46/47
PDPOSTIPS-G11	13.93; 3/23	27.71; 1/14	-16.39; 16/61	27.11; 0/9
PDPOSTIPS-G2	19.45; 4/16	23.00; 4/19	6.36; 27/31	2.09; 20/23
PDPOSTIPS-G6L11	9.67; 5/9	36.27; 0/4	5.03; 4/10	12.63; 0/2
PDPOSGSC-L2	20.65; 6/15	-2.64; 6/15	0.93; 27/46	-22.23; 24/33
PDPOSGSC-G2L3	17.31; 15/24	-14.81; 12/28	22.35; 49/47	-9.35; 40/55
PDPOSGSC-G3L6	20.57; 9/12	3.66; 8/20	-11.37; 20/38	-5.73; 17/20
PDPOSGSC-G6L7	0.95; 12/16	-10.15; 15/20	1.69; 49/43	5.19; 21/22
PDPOSGSC-G7L11	4.92; 11/25	1.05; 12/25	-20.33; 26/36	1.88; 26/32

The complete original panel CSV also preserves every original regression candidate, selection result, native-unit mean error amount and failure status. This table adds no refit or parameter selection and does not extend the independent-increment population guarantee to these financial series.

S14 Earlier-period fixed selection rules

The earlier-period experiment applies the saved fixed selection rules to familiar historical data. These origins already have validation and evaluation roles in the current records, including actual rolling-fold scoring. They cannot now be treated as untouched evidence for the expanded backcast. The experiment remains retrospective, not external or prospective validation. R0 denotes the fixed minimum-validation-loss rule. R2 is the predesignated main one-standard-error selection rule, not a result-dependent replacement chosen for its subsequent performance. The rule design precedes its calculations but follows earlier research results.

R1 adds the ordinary empirical quantile to the finite regression candidates. R2 compares each candidate's validation losses with the same best-mean candidate, using the paired-difference Bartlett-HAC standard error. Candidates within that tolerance qualify, and the fixed rule prefers the strongest shrinkage, including the empirical report. This paired HAC tolerance is a time-series adaptation of a one-standard-error preference, not the textbook formula, a significance test or a noninferiority guarantee. It does not account for the whole uncertainty from selecting the best candidate on that segment.

R3 applies the same preference to prespecified earlier rolling folds and the outer validation segment, with origin-equal weighting and maturity gaps. Its HAC pairs use actual source-index distances rather than compressing gaps. Without an eligible earlier fold, the saved design makes it identical to R2; this is not a numerical fallback. Earlier-period results were sealed before the already-inspected-period calculations that could reuse those origins.

Table S12: Every earlier-period minimum-loss and one-standard-error comparison. The one-standard-error rule misses the recorded success conditions.

Series	Horizon / history	Minimum-loss rule (%)	One-standard-error rule (%)	Breaches minimum-loss / one-standard-error	Historical breaches
PDPOSGS-B	6/52	11.22	7.43	10/10	7
PDPOSGS-B	24/52	181.39	181.39	27/27	8
PDPOSMBS-TOT	6/52	-4.11	3.83	7/6	4
PDPOSMBS-TOT	24/52	48.55	26.55	14/13	10
PDPOSCS-TOT	6/52	-11.56	-11.56	0/0	0

Table S12: Continued.

Series	Horizon / history	Minimum-loss rule (%)	One-standard-error rule (%)	Breaches minimum-loss / one-standard-error	Historical breaches
PDPOSCS-TOT	24/52	-4.12	1.02	4/4	2
PDPOSFGS-TOT	6/52	-27.32	-27.32	6/6	0
PDPOSFGS-TOT	24/52	-3.87	-3.60	4/3	2
PDPOSTIPS-L2	6/52	-7.06	-7.06	2/2	1
PDPOSTIPS-L2	24/52	-23.33	-1.55	4/1	1
PDPOSTIPS-G11	6/52	-5.55	-5.79	5/5	4
PDPOSTIPS-G11	24/52	5.67	-0.87	15/15	15
PDPOSTIPS-G2	6/52	15.16	15.16	8/8	0
PDPOSTIPS-G2	24/52	30.27	30.27	14/14	10
PDPOSTIPS-G6L11	6/52	283.88	28.58	17/6	0
PDPOSTIPS-G6L11	24/52	-4.03	-2.70	2/1	0
PDPOSGSC-L2	6/52	48.04	16.09	17/15	14
PDPOSGSC-L2	24/52	40.77	40.77	34/34	30
PDPOSGSC-G2L3	6/52	1.40	11.14	12/13	8
PDPOSGSC-G2L3	24/52	-11.22	15.48	15/20	15
PDPOSGSC-G3L6	6/52	2.63	2.35	11/11	10
PDPOSGSC-G3L6	24/52	15.62	6.92	29/28	26
PDPOSGSC-G6L7	6/52	19.03	19.03	6/6	1
PDPOSGSC-G6L7	24/52	2.05	1.82	20/19	19
PDPOSGSC-G7L11	6/52	19.18	12.77	13/13	10
PDPOSGSC-G7L11	24/52	8.48	2.70	29/29	27
fx_hourly	6/720	2.44	0.62	79/73	63
fx_hourly	6/2880	-16.62	-6.28	20/20	16
fx_hourly	24/720	-4.36	-4.36	47/47	32
fx_hourly	24/2880	-1.42	-1.40	1/1	0

All three recorded success criteria are marked NOT_MET. R2's worst weekly excess is 181.39 percent, and its weekly median is 5.37 percent. The long-history FX gain also fails the prewritten retention criterion. This adverse result remains a robustness finding rather than a contribution claim. Complete inspected-period and all-rule rows remain linked in the R6 records.

S15 Selection failures and replication boundaries

Required validation-candidate failure blocks the dependent rule; it is neither retried nor replaced by a shorter mean. The complete inventory of non-complete reports is below. Every listed row has the saved reason `REQUIRED_VALIDATION_CANDIDATE_FAILED`. Those failures remain part of the evidence even though other rules in the same task can complete.

Table S13: All non-complete fixed-rule report rows. Failed mandatory candidates remain failures.

Phase	Series	Horizon / history	Rule	Status
retro	fx_hourly	6/720	R3	FAILED
retro	fx_hourly	24/720	R3	FAILED
inspected	fx_hourly	6/720	R3	FAILED
inspected	fx_hourly	24/720	R3	FAILED

The Treasury earlier-period structural comparison uses a shorter history than the original primary. It is explicitly an analogue, not an exact earlier replication of the long-history primary. The source-reported comparison and its definition are preserved in the primary-structure table. All original primary values remain unchanged. Reclassifying the analogue as an exact primary replication, or calling familiar retrospective FX data an unseen dealer, would change the scientific claim without adding evidence.

S16 Finite delayed calibration and headroom SAV

The finite delayed ACI-style wrapper adapts to past strict hits after their entire peak labels mature. Its score pool contains only completed residual scores and uses the specified finite memory. The base-plus-score quantile is projected onto the known floor. The adaptive alpha state is projected to the current feasible range; the rank-zero convention issues the floor. Gamma was selected by coverage distance, then pinball loss and the fixed tie rule, a coverage-first selection criterion distinct from original regression fitting. No canonical ACI theorem is imported to this modified procedure.

The headroom SAV method adapts a CAViaR recursion, rather than replicating a return-VaR model. The completed headroom response is normalized by its training-only scale; the recursive report is multiplied back by that same scale and then floor-projected. Its fixed bounded optimizer and starts retain all eligibility and convergence failures; a complete fit is not a proof of a global optimum. Selection state carries across the unissued chronological gap.

Table S14: Complete delayed-wrapper and headroom-recursion inventory. Two long-horizon wrapper cells remain unavailable.

Source	Horizon / history	Method	Mean loss	Breaches	Status
FX	6/720	ACI_F	472677.77	331	COMPLETE
FX	6/720	ACI_QR	439672.91	352	COMPLETE
FX	6/720	SAV	449237.37	349	COMPLETE
FX	6/2880	ACI_F	463271.47	328	COMPLETE
FX	6/2880	ACI_QR	393795.06	329	COMPLETE
FX	6/2880	SAV	456696.22	364	COMPLETE
FX	24/720	ACI_F	921402.08	322	COMPLETE
FX	24/720	ACI_QR	1020220.09	393	COMPLETE
FX	24/720	SAV	781586.36	439	COMPLETE
FX	24/2880	ACI_F	791441.18	323	COMPLETE
FX	24/2880	ACI_QR	755982.70	294	COMPLETE
FX	24/2880	SAV	756811.00	333	COMPLETE
Bills	6/52	ACI_F	3818.40	18	COMPLETE
Bills	6/52	ACI_QR	3489.82	29	COMPLETE
Bills	6/52	SAV	2974.32	16	COMPLETE
Bills	6/260	ACI_F	2792.91	7	COMPLETE
Bills	6/260	ACI_QR	3013.16	6	COMPLETE
Bills	6/260	SAV	2892.23	29	COMPLETE
Bills	24/52	ACI_F	4368.10	38	COMPLETE
Bills	24/52	ACI_QR	6250.53	47	COMPLETE
Bills	24/52	SAV	4182.40	35	COMPLETE
Bills	24/260	ACI_F	NA	NA	NA
Bills	24/260	ACI_QR	NA	NA	NA
Bills	24/260	SAV	3100.97	25	COMPLETE

The NA wrapper rows have the saved reason `VALIDATION_INELIGIBLE`; no test is substituted for the absent forecasts. All losses are in each source's native units and must not be pooled.

S17 Wrapper comparisons, uncertainty and missing cells

Comparing wrapped regression with wrapped historical reports is a different question from comparing either wrapper with its unwrapped base. The following table retains every within-cell comparison. A lower coverage distance measures closeness of the empirical hit rate to the target probability, not conditional coverage or lower monetary loss.

Table S15: Every paired wrapper comparison. Calibration adjustment preserves the long-history short-horizon FX score gain but does not produce universal dominance.

Source	Horizon / history	Regression vs historical (%)	Breaches regression / historical	Regression adjustment	Historical adjustment
FX	6/720	-6.98	352/331	0.00313	0.00004
FX	6/2880	-15.00	329/328	0.00034	0.00049
FX	24/720	10.72	393/322	0.00932	0.00140
FX	24/2880	-4.48	294/323	0.00562	0.00125
Bills	6/52	-8.61	29/18	0.05469	0.01498
Bills	6/260	7.89	6/7	0.02834	0.02473
Bills	24/52	43.09	47/38	0.11968	0.08718
Bills	24/260	NA	NA	NA	NA

The working-reference comparison inventory contains 56 fixed slots, with 5 raw p values unavailable. The reported Holm bookkeeping keeps those slots; mapping an NA component to a bookkeeping value does not create an observed p value or validate failed reference assumptions. HLN references remain separately identified. Exact update equations, gamma candidates, selection freezes, numerical receipts and all comparisons remain in the v6 replay guide and the complete WP2 inventory. The original primaries are not overwritten by these post-result / exploratory extensions. The upper-horizon NA rows and adverse SAV comparisons remain visible rather than becoming a favorable subset.

S18 Untuned forecast combinations

The median combination lowers Treasury-bill loss by 4.0%, while the mean combination raises it by 2.8%.

Fixed report combinations use the matched historical and original regression reports at identical origins. Their arithmetic mean is untuned. The median also requires the original selected plain local-quantile report; neither a smaller predictor model nor the endpoint plug-in substitutes for that missing constituent. Combining floor-respecting reports preserves the floor, but does not guarantee a conditional quantile or a score below that of the best constituent. Identical reports generate an NA equal-loss p value rather than a manufactured significance result.

Table S16: All original-cell combination comparisons. A report combination can outperform one constituent while losing to the other.

Source	Horizon / history	Rule	vs historical (%)	vs regression (%)	Breaches
FX	6/720	Mean	-6.91	-4.90	338
FX	6/720	Median	-4.81	-2.75	325
FX	6/2880	Mean	-10.61	4.87	330
FX	6/2880	Median	-11.17	4.22	327
FX	24/720	Mean	5.50	-19.77	533
FX	24/720	Median	5.00	-20.15	514

Table S16: Continued.

Source	Horizon / history	Rule	vs historical (%)	vs regression (%)	Breaches
FX	24/2880	Mean	-8.94	-5.39	345
FX	24/2880	Median	-3.74	0.01	416
Bills	6/52	Mean	-3.15	-47.18	26
Bills	6/52	Median	0.00	-45.46	11
Bills	6/260	Mean	2.77	-31.84	40
Bills	6/260	Median	-3.98	-36.32	28
Bills	24/52	Mean	18.87	-63.69	54
Bills	24/52	Median	0.00	-69.46	33
Bills	24/260	Mean	-5.32	-54.88	28
Bills	24/260	Median	-11.51	-57.83	14

Table S17: Complete combination counts including the added panel. Missing median constituents remain NA.

Rule	Total	Complete	Unavailable	Lower loss than both	Lower loss than historical	Lower loss than regression	Equal loss to historical
Mean	56	56	0	14	16	54	0
Median	56	8	48	3	5	6	2

The added panel lacks the specified plain local-quantile constituent, so all its median rows remain unavailable. Mean combinations are complete, but lower loss than regression alone does not imply lower loss than the historical rule. All panel-specific values and every comparison remain in the complete combination record.

S19 Combination Murphy diagrams and comparison families

Elementary-score comparisons vary the score threshold while keeping the quantile level fixed. Exact integer-event signs at every unique breakpoint determine sample crossing before display interpolation. A picture is not used to select favorable thresholds. These matched combination contexts are distinct from the original FX primary with its shorter rolling reference. Table S18 reports every combination's elementary-score crossing diagnostic.

Table S18: Every combination Murphy comparison in the original-primary contexts. Crossing prevents threshold-uniform sample dominance.

Source	Left-hand method	Right-hand method	Ranking	Positive breakpoints	Negative breakpoints
FX	MEAN_F_QR	F	CROSSING	217	15628
FX	MEAN_F_QR	QR	CROSSING	13562	2343
FX	MEDIAN_F_QR_KNN	F	CROSSING	909	13572
FX	MEDIAN_F_QR_KNN	QR	CROSSING	10827	2575
Bills	MEAN_F_QR	F	CROSSING	336	291
Bills	MEAN_F_QR	QR	CROSSING	179	454
Bills	MEDIAN_F_QR_KNN	F	CROSSING	262	274
Bills	MEDIAN_F_QR_KNN	QR	CROSSING	147	446

The complete normal-DM comparison family has 224 slots, of which 98 have NA raw p values. The family adjustment is bookkeeping for those saved working references, not error control over the com-

plete historical analysis search. The inspected record, overlapping labels, repeated fitting and related aggregate series remain common to the methods. All original and adverse rows remain in the combination findings. Exact breakpoint signs support only the stated finite-sample score comparison; they do not certify general forecast dominance, optimal funding, or a causal benefit from acting on the limits.

S20 Training-defined position-change states

The states are low, middle and high issue-known RMS position-change states, not price volatility or yield regimes. Cutpoints come only from the training history. Reports and original primary means stay unchanged; the diagnostic divides the already viewed evaluation record without estimating a switching policy. Source-native group differences are not pooled.

Table S19: All three state groups for each original primary. The FX regression-minus-reference loss difference remains negative in every group, while the weekly evidence remains adverse.

Source	State	Issues	Regression minus reference loss	Dependence-adjusted standard error
FX	low	1593	-58761.27	16492.94
FX	middle	1816	-27839.43	16257.70
FX	high	3216	-70387.91	17895.89
Bills	low	9	6696.66	1705.21
Bills	middle	25	5230.59	2608.15
Bills	high	243	810.88	466.48

Table S20: Complete state diagnostic reference family. A joint rejection need not identify a useful state rule.

Source	Diagnostic	Unadjusted p-value	Holm-adjusted p-value	Status
FX	joint_GW	1.46298e-06	5.8519e-06	OK
FX	group_heterogeneity	0.159383	0.159383	OK
Bills	joint_GW	0.0484878	0.0969756	OK
Bills	group_heterogeneity	0.000426772	0.00128032	OK

These finite-instrument and group-mean HAC references require unestablished moment, dependence and covariance assumptions. A joint conditional-moment rejection can be driven by an unconditional mean difference; it is not evidence that a state-dependent trading or limit-selection policy works. The source-level reconstruction and all cutpoints remain in the complete state record.

S21 Refitted target-level sensitivities

Changing the target quantile requires fitting the specified target-level procedure; it is different from reweighting the errors of a fixed report. The following saved sensitivities keep the original horizon and reference structure. The weekly primary and matched reference coincide and count once. All unavailable upper-target rows remain NA with `VALIDATION_INELIGIBLE`, rather than being replaced by a lower target or a favorable subset. Absolute loss values across different quantile levels measure different objectives.

Table S21: Complete target-level sensitivity. The lower-target directions recur, but upper-target evidence remains unavailable.

Source	Target τ	Reference	Mean loss difference	Regression breaches	Reference breaches	Status
FX	0.90	primary	-60715.44	686	620	OK
FX	0.90	matched	-67628.17	686	667	OK
Bills	0.90	primary	1175.03	73	55	OK
FX	0.99	primary	NA	NA	NA	NA
FX	0.99	matched	NA	NA	NA	NA
Bills	0.99	primary	NA	NA	NA	NA

The complete family contains 6 slots, with 3 raw p values unavailable. The fixed target-specific selections precede their evaluation; full settings and results remain in the target-level inventory. The saved lower-target directions do not establish robustness to an arbitrary tail level, actual shortage costs, or a new economic decision rule. Failed eligibility is a result of the specified procedure and remains part of the paper's boundary.

S22 Horizon coherence and late conditional-hit evidence

Realized peaks are ordered as the forecast horizon grows, but independently fitted reports use horizon-specific mature peak responses and can cross. The following exact strict comparisons keep all common issue origins. Repairing a long-horizon report upward can worsen pinball loss; target ordering alone does not identify a universally beneficial repair. Tables S22–S23 retain horizon-coherence results and the complete late diagnostic.

Table S22: Every original quantile-regression horizon-coherence comparison. Ordered targets do not force ordered fitted reports.

Source	History	Issues	Forecast crossings	Target violations
FX	720	6625	640	0
FX	2880	6625	140	0
Bills	52	277	91	0
Bills	260	277	80	0

The late fixed-terminal conditional-hit diagnostic uses the null $E[I_t \mid \mathcal{F}_t] \leq \alpha$, where $I_t = 1\{P_t > C_t\}$. Within each globally anchored maturity phase, a fixed upper-probability alternative supplies hit/no-hit factors with conditional expectation at most one under the stated null. The procedure mixes whole products over its fixed prior, then averages dependent phases. It does not multiply phases or select their maximum. The fixed-family threshold does not restore retrospective error control after viewed-sample diagnostic selection.

Table S23: Every threshold-crossing cell in the complete late diagnostic. A noncrossing elsewhere is not a calibration certificate.

Source	Horizon / history	Breaches / issues	Saved display value
FX	24/720	924/6625	8.499112682788E+7
Bills	6/52	84/277	1.310958812771E+8
Bills	6/260	54/277	2.204375927802E+3
Bills	24/52	132/277	3.814063640027E+4

The complete summary and phase tables, exact rational identities, all noncrossings and the interpretation boundaries remain in the e-value records. This is a fixed-terminal diagnostic, not an anytime p value, optional-stopping guarantee, new estimator, or independent validation. Score, breach amount, frequency and conditional-hit evidence answer different questions.

S23 Numerical implementation and inference conventions

Completed labels and update clocks. At a scheduled refit κ, the latest mature peak response origin is $\kappa - H$. A history of W mature peak responses therefore uses $\{\kappa - H - W + 1, \ldots, \kappa - H\}$, with every target ending by that refit. Training transforms, scales, coefficients and historical pools are held until the next scheduled refit. The current observed position may still change at each forecast issue. Replacing that issue anchor by the refit anchor changes the forecasting procedure.

Regression and local quantiles. The regression objective uses a training-normalized headroom response, an unpenalized intercept and a ridge penalty on slopes. Projection is applied when issuing the report and evaluating its validation score, not inside the fitted training objective. Local methods use training-only standardized predictors, fixed neighbour menus and duplicate-preserving order statistics. A selected whole-pool local quantile is an exact historical identity, not a failed-fit fallback. Scale-adjusted neighbours change responses as well as retrieval.

Numerical eligibility. The accepted solver records primal and dual residuals, objectives, feasibility and the gap between the primal and dual objectives. Ill-conditioned, nonfinite, unsupported-rank and failed candidate states retain their original statuses under each named selection contract. A mandatory-candidate failure blocks a procedure when its contract requires every candidate. The original full regression has that requirement; the four-feature selector may choose among completed eligible candidates. A failed selected evaluation never receives a replacement penalty or a shortened score. Changing the ridge menu changes the procedure even when feature definitions stay fixed. A numerical certificate does not establish financial accuracy, economic relevance, inferential validity or production continuity. No production continuity or fallback policy is evaluated here.

Working-reference inference. Saved unconditional loss differences use the recorded Bartlett-HAC convention; HLN and doubled-bandwidth sensitivities are labeled separately. Separate families and later reporting partitions do not erase the wider selection history. Coverage references distinguish full-grid Kupiec from the transition-conditional Christoffersen joint likelihood, which conditions on the initial hit. Small sparse grids and boundary likelihoods retain their original status. Neither a nonrejection nor a small working-reference p value proves conditional coverage or supplies selection-adjusted confirmation.

Common-pool construction distances. On one nonempty historical episode set, compare the historical headroom, translated-path and historical-peak reports using the same available order statistic. Each episode peak includes its starting observation. A translated path changes each level by the same anchor difference, so its absolute peak changes by at most that difference. Flooring is nonexpansive and commutes with the fixed order statistic. The headroom construction has the corresponding bound in the difference of anchor magnitudes. Thus

$$|C^{\mathrm{headroom}} - C^{\mathrm{peak}}| \le \max_i \big||N_i| - B\big|, \qquad |C^{\mathrm{translated}} - C^{\mathrm{peak}}| \le \max_i |N_i - x|.$$

For a proof of the order-statistic step, paired coordinatewise differences bounded by a radius imply that each empirical quantile lies below the other plus that radius; reverse the roles for the second inequality. These deterministic same-pool bounds do not order financial accuracy or coverage. Changing historical rows on one side invalidates the common-pool argument.

Exact menus, numerical tolerances, failures and diagnostic tables remain in EXTENDED RESULTS and the dated source records.

S24 Fixed controlled comparison of inherited procedures

The simulation compares inherited reports for mean-reverting position levels, not positively autocorrelated increments and not a fitted reproduction of either financial source. It uses

$$N_t = \phi N_{t-1} + \varepsilon_t, \qquad \varepsilon_t = \sqrt{v_t} z_t, \qquad v_t = \omega + a\varepsilon_{t-1}^2 + b v_{t-1}, \qquad z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

The level persistences are $\phi \in \{0.7, 0.9, 0.98\}$, variance persistences $p = a + b \in \{0, 0.9, 0.98\}$, histories $W \in \{52, 260, 720, 2880\}$, and horizons $H \in \{6, 24\}$. For the increasing variance-persistence settings, the (ω, a, b) triples are $(1, 0, 0)$, $(0.1, 0.05, 0.85)$, $(0.02, 0.05, 0.93)$. Each innovation has unit unconditional variance under the specified recursion. Initial level and variance are fixed; a burn-in of 20000 observations precedes 3264 retained observations. This is a long-burn approximation, not an exact stationary initialization. The common hourly calendar has no simulated calendar effect.

There are 32 independent repetition streams, with master entropy 2026102408 and common normal innovations across conditions within a repetition. Validation uses origins 3072–3143; evaluation uses 3168–3239, with 72 overlapping issues and a refit every 24 records. Repetition, not issue or grid cell, is the Monte Carlo uncertainty unit. Training uses exactly W origins $r - H - W + 1, \ldots, r - H$, with every label ending by refit r. The current-position floor and query features use the actual issue state. The full model keeps its original predictors and the four-feature model its original four; their penalty menus and failure rules differ, so their contrast is not an isolated intervention on predictor count.

For each cell, let A_j and B_j be regression and matched-historical mean losses in repetition j, and R the planned number of repetitions. The reported relative change and its delta-method Monte Carlo standard error are

$$\widehat{\Delta} = 100\left(\frac{\overline{A}}{\overline{B}} - 1\right), \qquad \mathrm{MCSE}(\widehat{\Delta}) = \frac{100\, s(A_j - \widehat{r} B_j)}{\sqrt{R}\,\overline{B}}, \qquad \widehat{r} = \frac{\overline{A}}{\overline{B}}.$$

Sample standard deviations use divisor $R - 1$. Breach-rate errors use the between-repetition standard deviation divided by $\sqrt{R}$. Strict wins require $A_j < B_j$, with error $\sqrt{\widehat{\pi}(1-\widehat{\pi})/R}$. At zero or all wins that plug-in error is zero without implying certainty; the worst-case error bound is $0.5/\sqrt{R}$, or 8.84 percentage points here. The short evaluation segments, common innovations across cells, and fixed process family limit a regime claim.

The fixed descriptive clear-gain screen requires all 32 paired repetitions, $\widehat{\Delta} + 2\,\mathrm{MCSE}(\widehat{\Delta}) < 0$, and a strict-win fraction of at least 0.75. Its target is $H = 6$, $W \geq 720$, $p \geq 0.9$. Support additionally requires at least 6 of 12 clear-gain target cells and none among the 60 outside cells. Each regression has only 2 target gains and 6 outside gains. The recorded conclusion is **MISMATCH**: both the target count and the no-outside requirement fail. This screen has no family-wide significance guarantee and supplies no new population endpoint-bound guarantee under dependent financial paths.

The complete two-regression 72-cell tables remain unchanged in v9 Tables S26–S27 and their native fields in the saved controlled simulation. All adverse, failed and unavailable statuses retain their original meanings. The preceding design and prediction outcomes refer to that entire grid, not a successful subset.

Candidate and procedure statuses. The complete inventory contains 6912 complete method tasks, 18431 complete validation candidates and 1 failed candidate. The failure is the four-feature candidate at zero-based repetition 23, $\phi = 0.9$, $p = 0$, $W = 52$, $H = 24$, and penalty 0.1. Its duality-gap criterion fails at 10000 iterations. The fixed eligible-completed-candidate rule selects penalty 0.01, whose full evaluation completes; this is not a retry or substitution of a failed selected evaluation. The original full-regression selector instead requires every declared validation candidate. Incomplete procedures retain NA full-procedure performance, with any available-pair summary labeled conditional on completion. The saved source summaries retain all cells and their full precision in `results/revision-v8/wp3/`

summary.json; all six loss, breach and win heatmaps remain alongside them. Both complete numerical grids retain every cell.

S25 Same-dealer completed-day aggregation

The pinned hourly FX snapshot produces 930 nonempty source-naive calendar days under the fixed last-observation rule, rather than the approximate sample size in the planning instruction. Each report is issued only at the end of its completed source-naive day; the selected row's original timestamp is not treated as an earlier time at which the identity of the last row was already known. Missing days stay absent, and partial boundary days remain. The timezone and historical first-release availability are unknown. This is a last-record proxy, not a verified financial close or a peak over intraday excursions.

Histories and horizons count subsequent nonempty daily observations. The split copies the Treasury observation-index geometry, not its absolute calendar dates or a proportional split: validation runs from 2015-02-24 to 2015-04-24 and evaluation from 2015-05-24 to 2016-12-27 on 501 common issues. Raw or resampled positions and per-issue trajectories are not redistributed.

Table S24 reports every same-dealer daily comparison.

Table S24: Same-dealer daily sampled-position results; history and horizon count recorded nonempty calendar days. All methods share the declared evaluation grid; loss is scaled to USD millions for display only. The four-feature report yielding the isolated score gain has more breaches than its historical reference.

History	Horizon	Procedure	Penalty	Mean loss (USD m)	Loss change vs historical (%)	Breaches count / rate (%)
52	6	Historical headroom	–	1.049	–	36 / 7.186
52	6	33-feature regression	0.1	2.032	93.586	146 / 29.142
52	6	Four-feature regression	0.001	1.974	88.128	129 / 25.749
52	24	Historical headroom	–	1.553	–	66 / 13.174
52	24	33-feature regression	0.01	3.456	122.589	206 / 41.118
52	24	Four-feature regression	0.01	3.163	103.719	225 / 44.910
260	6	Historical headroom	–	0.964	–	30 / 5.988
260	6	33-feature regression	0.1	1.175	21.873	54 / 10.778
260	6	Four-feature regression	0.1	0.935	-2.974	37 / 7.385
260	24	Historical headroom	–	1.882	–	24 / 4.790
260	24	33-feature regression	0.001	2.642	40.335	99 / 19.760
260	24	Four-feature regression	0.001	2.177	15.627	46 / 9.182

Historical headroom wins seven of the eight named comparisons, but the four-feature longer-history, shorter-horizon daily exception lowers loss by about 3.0% with 37 regression breaches versus 30 historical breaches. Both regressions meet the fixed majority-based descriptive classification, not a dominance or coverage theorem. The recorded source identity is held fixed while physical history, horizon, update cadence, calendar predictors, sampled response and evaluation span change jointly. The contrast cannot identify a pure frequency effect or supply independent dealer validation.

S26 Fixed-parameter full-prefix backcast and overlapping windows

The additional retrospective backcast uses the complete eligible prefix 3071–10438, with 7368 common issue origins before the original later validation. It retains the original later-selected parameters: full regression: $H = 6, \lambda = 0.01$, $H = 24, \lambda = 0.001$; four-feature regression: $H = 6, \lambda = 0.01$, $H = 24, \lambda = 0.01$. Coefficients, transformations and training labels obey $i + H \leq r$ at each refit, but those fixed penalties were chosen after the historical issue dates. Thus this is a fixed-parameter pseudo-out-of-sample backcast with later-selected parameters, not a real-time historical validation.

The full prefix contains all 1440 earlier reported backcast origins, with inclusive issue indices 8999–10438. Actual earlier rolling-validation targets at the longer horizon cover 3071–10462, exactly the new prefix's longer-horizon target range. Apparent issue gaps between old validation blocks are not untouched target paths. Three displayed overlapping/inspected windows therefore do not create a fresh independent third validation period. This **MISMATCH** remains regardless of the direction of the added score comparisons. No window effects, p-values or success counts are pooled.

Full-prefix regression lowers matched loss by 12.2% but records 416 versus 282 breaches. Its rolling-reference DM p-value is 0.055. Longer-horizon gains coexist with reference-sensitive losses in the complete tables.

Table S25: Six-step original regression across three inspected windows; the full-prefix window contains the earlier reported backcast, so this is not independent validation. Loss is scaled to USD millions for display; raw normal-HAC p-values are descriptive. Missing saved breach counts remain NA.

Window	Reference	Mean loss Reg. / hist. (USD m)	Change (%)	Breaches Reg. / hist.	DM	Raw p	Issues
Full prefix	Matched F2880	0.587 / 0.668	-12.167	416 / 282	-3.518	<0.001	7368
Full prefix	Rolling F720	0.587 / 0.619	-5.296	416 / 387	-1.918	0.055	7368
Prior backcast	Matched F2880	0.287 / 0.344	-16.624	20 / 16	-4.178	<0.001	1440
Prior backcast	Rolling F720	0.287 / 0.278	3.168	20 / NA	0.563	0.574	1440
Original	Matched F2880	0.386 / 0.453	-14.767	376 / 360	-5.866	<0.001	6625
Original	Rolling F720	0.386 / 0.442	-12.660	376 / 332	-5.182	<0.001	6625

Table S26: Complete full-prefix backcast comparisons; the two regressions use the original later-selected parameters. All planned comparisons are shown; raw normal-HAC p-values are descriptive. The window is already inspected and overlaps the earlier reported backcast.

Horizon	Regression	Reference	Mean loss Reg. / hist. (USD m)	Change (%)	Breaches Reg. / hist.	DM (raw p)
6	33-feature	Matched F2880	0.587 / 0.668	-12.167	416 / 282	-3.518 (<0.001)
6	33-feature	Rolling F720	0.587 / 0.619	-5.296	416 / 387	-1.918 (0.055)
6	Four-feature	Matched F2880	0.617 / 0.668	-7.620	290 / 282	-2.498 (0.012)
6	Four-feature	Rolling F720	0.617 / 0.619	-0.393	290 / 387	-0.177 (0.859)
24	33-feature	Matched F2880	1.470 / 1.783	-17.538	470 / 275	-1.997 (0.046)
24	33-feature	Rolling F720	1.470 / 1.431	2.730	470 / 477	0.315 (0.752)
24	Four-feature	Matched F2880	1.408 / 1.783	-21.027	355 / 275	-2.778 (0.005)
24	Four-feature	Rolling F720	1.408 / 1.431	-1.617	355 / 477	-0.228 (0.820)

The fixed short-horizon full-regression descriptive screen retains matched score gains across the three inspected windows and a changing rolling-reference direction. It is not confirmation after inspection or independent replication. The complete longer-horizon results also retain matched score gains, and the full regression's loss against the rolling reference. A score gain can coexist with more breaches; neither a conditional-coverage certificate nor an economic operating benefit follows from these tables.

The added eight pairwise tests use only raw standard-normal two-sided DM-type working references, separate from all original adjusted families. For n paired losses at horizon H,

$$L = \min\bigl(n-1, \max\bigl\{H, \lfloor 4(n/100)^{2/9} \rfloor\bigr\}\bigr)\,.$$

The centered Bartlett Newey–West covariance uses divisor n, and the statistic is the mean regression-minus-reference loss divided by $\sqrt{\widehat{\mathrm{LRV}}/n}$. Nonpositive long-run variance or constant loss differences retain an unavailable statistic and their specified status; there is no epsilon repair. These p-values receive no multiplicity or full selection adjustment and do not establish equivalence, real-time validity

or independent confirmation. Missing old longer-horizon original-menu, four-feature, rolling-DM or breach-count fields stay NA; the old five-menu rule is not relabeled as the original three-menu regression. No missing old test was recalculated.

S27 Earlier calibrated simulation and numerical failures

This distinct calibrated design uses constant-variance AR(1) levels, where the peak depends on the current position (D0), an hourly AR–GARCH–t law with a known hour shape (D1), and a weekly AR–GARCH–t law (D2). D0/D2 share the original pre-validation Treasury calibration; D1 uses the original pre-validation FX prefix. Levels are divided by the prefix standard deviation without demeaning. The fixed calibration, clocks, origin geometry, parameters and numerical rules are retained in the imported calibrated records. The weekly clock, validation span, refit schedule and six-candidate menu, including the intercept-only limit, differ from Section S24. Their short-history outcomes cannot be treated as replications of one design.

Each completed cell reports repetition medians and quartiles of the within-repetition relative loss against F. These differ from the ratio of across-repetition mean losses in Section S24. Quartiles describe dispersion under fixed fitted laws, not confidence limits. The known-state oracle is a finite future-path approximation; its realized score need not beat F in every repetition. Ordinary-rank I, oracle results, strict breaches and intercept-only selections remain separate diagnostics.

Table S27: Legacy calibrated simulation: all 24 planned cells.

Process	History	Features	Complete / required	Regression Δ [IQR]	Win	Intercept change	Oracle change	Historical hit	Intercept hit	Regression hit	Oracle hit	Infinite penalty
D0	52	4	100/100	$-0.2\,[-4.6, 22.1]$	0.510	+0.055	-30.0	0.046	0.064	0.129	0.048	0.110
D0	52	33	100/100	$1.1\,[-2.0, 6.1]$	0.390	+0.055	-30.0	0.046	0.064	0.094	0.048	0.300
D0	260	4	100/100	$-15.0\,[-20.4, -7.3]$	0.980	+0.030	-24.6	0.050	0.056	0.068	0.048	0.000
D0	260	33	100/100	$-2.2\,[-8.2, 2.7]$	0.640	+0.030	-24.6	0.050	0.056	0.094	0.048	0.060
D0	720	4	100/100	$-18.8\,[-24.4, -14.5]$	0.980	+0.046	-23.3	0.050	0.051	0.055	0.048	0.030
D0	720	33	100/100	$-14.6\,[-21.0, -9.7]$	0.980	+0.046	-23.3	0.050	0.051	0.064	0.048	0.050
D0	2880	4	100/100	$-21.8\,[-25.2, -17.2]$	0.990	-0.005	-23.6	0.050	0.050	0.048	0.048	0.020
D0	2880	33	100/100	$-19.8\,[-23.7, -15.2]$	0.990	-0.005	-23.6	0.050	0.050	0.052	0.048	0.020
D1	52	4	10/100	**FAIL**	–	–	–	–	–	–	–	–
D1	52	33	100/100	$0.2\,[-1.1, 2.1]$	0.470	+0.013	-34.0	0.083	0.096	0.096	0.055	0.770
D1	260	4	4/100	**FAIL**	–	–	–	–	–	–	–	–
D1	260	33	56/100	**FAIL**	–	–	–	–	–	–	–	–
D1	720	4	3/100	**FAIL**	–	–	–	–	–	–	–	–
D1	720	33	17/100	**FAIL**	–	–	–	–	–	–	–	–
D1	2880	4	3/100	**FAIL**	–	–	–	–	–	–	–	–
D1	2880	33	32/100	**FAIL**	–	–	–	–	–	–	–	–
D2	52	4	100/100	$1.2\,[-3.1, 19.8]$	0.430	-0.485	-29.5	0.055	0.070	0.101	0.046	0.180
D2	52	33	100/100	$0.4\,[-1.6, 5.7]$	0.460	-0.485	-29.5	0.055	0.070	0.098	0.046	0.270
D2	260	4	100/100	$-13.9\,[-20.5, -5.2]$	0.910	-0.036	-25.6	0.052	0.056	0.069	0.046	0.030
D2	260	33	100/100	$-0.8\,[-7.8, 2.9]$	0.600	-0.036	-25.6	0.052	0.056	0.097	0.046	0.080
D2	720	4	100/100	$-18.6\,[-22.6, -10.9]$	0.980	-0.044	-24.8	0.046	0.048	0.054	0.046	0.030
D2	720	33	100/100	$-12.4\,[-18.8, -4.1]$	0.910	-0.044	-24.8	0.046	0.048	0.059	0.046	0.050
D2	2880	4	100/100	$-20.1\,[-24.9, -15.3]$	0.970	-0.017	-24.1	0.044	0.046	0.047	0.046	0.030
D2	2880	33	100/100	$-20.6\,[-24.1, -14.4]$	0.950	-0.017	-24.1	0.044	0.046	0.052	0.046	0.080

Δ is the median relative loss change against F (percent); IQR is across pre-specified repetitions. Win and hit columns are proportions; I is the ordinary empirical quantile and O is the finite nested oracle. The ∞ column is the frequency of selecting the intercept limit. All seven failed D1 rows remain unavailable after ADMM nonconvergence, even when some repetitions completed. D0: constant-variance AR(1) levels, where the peak depends on the current position. D1: FX-calibrated hourly; D2: weekly-calibrated. The broader legacy menu includes the intercept limit; these results are not the WP3 ratio-of-mean-losses.

Table S28: First recorded failure in each unavailable legacy cell.

Process	History	Features	Rep	Stage	λ	Saved reason
D1	2880	4	1	evaluation	0.001	QuantileFitError:NONCONVERGENCE
D1	720	4	3	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	260	4	4	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	52	4	10	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	720	33	17	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	2880	33	32	validation	0.001	QuantileFitError:NONCONVERGENCE
D1	260	33	56	validation	0.001	QuantileFitError:NONCONVERGENCE

Repetition identifiers are zero based. Each event belongs to a whole failed cell in Table S27; no partial repetition summary is substituted. The original failure stage, penalty and reason are retained without a rerun.

A required numerical failure blocks the whole cell. Partial repetitions do not supply a replacement median, quartile, win rate or breach rate. The retained nonconvergence exceptions mean that the unchanged solver did not obtain its joint residual-and-gap acceptance certificate within its iteration budget. They establish neither nonexistence of an optimum nor poor predictive accuracy of an unavailable report. Final residuals and gaps were not retained; the four-feature exceptions also lack exact refit context. These quantities remain unknown. The original constant-variance and short-weekly predictions are unsupported; the hourly and history-order predictions are unassessable. These outcomes identify neither the financial loss mechanism nor a pure frequency effect.

S28 Pre-evaluation stability screen and switching stress simulation

The separately fixed stability-study protocol uses only pre-evaluation prefixes. Its joint OLS score includes intercept, AR slope and centered residual-square channels (Hansen, 1992). Each qualified whole-prefix AR–GARCH–t fit generates a plug-in parametric bootstrap reference. OLS and the score statistic are re-estimated on each bootstrap path; the generator is fitted once. Published iid critical values are not used. With the pre-specified $B = 999$ bootstrap paths, the p value is $(1 + \#\{H_b \geq H_{\text{obs}}\})/(B+1)$; the pre-specified draw count and all exceedances remain in the records. The table retains the observed statistic, bootstrap p value, Holm adjustment, moment qualification and descriptive cuts. Qualified fitted nulls require $\nu > 4$ and $q_4 = \alpha^2\{3(\nu-2)/(\nu-4)\} + 2\alpha\beta + \beta^2 < 1 - 10^{-8}$. A successful fit and this fourth-moment gate do not prove that the real law has those moments. The plug-in calibration supplies no exact finite-sample size guarantee; rejection can also reflect misspecification or outliers.

Table S29: Pre-specified prefix stability diagnostics: all 15 source rows.

Source	Statistic	p	Holm	Moment valid	q_4	Cuts	Dates	Segments	h	Break cap	State
GS-B	1.83	0.051	0.306	Y	0.905	104/293	2015-04-08/2018-11-21	104/189/111	104	Y	OK
MBS-TOT	1.26	0.208	0.5	Y	0.948	189/293	2016-11-23/2018-11-21	189/104/111	104	Y	OK
CS-TOT	1.18	0.022	0.2	Y	0.0642	none	–	404	104	N	OK
FGS-TOT	4.07	0.001	0.013	Y	0.981	138	2015-12-02	138/266	104	N	OK
TIPS-L2	1.13	0.02	0.2	Y	0.983	107	2015-04-29	107/297	104	N	OK
TIPS-G11	0.901	0.063	0.315	Y	0.972	none	–	404	104	N	OK
TIPS-G2	1.28	0.018	0.198	Y	0.167	none	–	404	104	N	OK
TIPS-G6L11	2.51	0.001	0.013	Y	0.973	163	2016-05-25	163/241	104	N	OK
GSC-L2	1.15	0.213	0.5	Y	0.976	289	2018-10-24	289/115	104	N	OK
GSC-G2L3	2.09	0.155	0.5	Y	0.978	104/269	2015-04-08/2018-06-06	104/165/135	104	Y	OK
GSC-G3L6	2.78	0.024	0.2	Y	0.94	117/288	2015-07-08/2018-10-17	117/171/116	104	Y	OK
GSC-G6L7	0.799	0.125	0.5	Y	0.926	none	–	404	104	N	OK
GSC-G7L11	1.64	0.024	0.2	Y	0.937	145	2016-01-20	145/259	104	N	OK

Continued on next page

Source	Statistic	p	Holm	Moment valid	q_4	Cuts	Dates	Segments	h	Break cap	State
FX daily	3.13	–	–	N	–	128/271	2014-07-06/2014-12-19	128/143/133	104	Y	NR
FX hourly	–	–	–	–	–	2036/5365	–	2036/3329/2743	1622	Y	OF

H: Hansen joint statistic; h: minimum segment transitions; M4: pre-specified fitted-null fourth-moment qualification. Cuts use transition-row units; dates mark the first target after each cut. Unarchived dates remain – even when cuts are available. Source codes omit the common PDPOS prefix. NR: daily FX NONREGULAR_NULL (Student fourth moment unavailable); OF: hourly FX GARCH_OPTIMIZATION_FAILED. BF: a restricted-boundary fit; OK does not certify the model. Unavailable raw p and Holm values remain –; internal Holm placeholders are not tests. Segmentation is descriptive, and the two-break cap does not establish significant individual breaks.

Candidate cuts minimize a separate-variance Gaussian AR quasi-likelihood criterion with penalty $(3 + 4k) \log M$, where M counts transitions. The fixed search allows at most two cuts and requires each segment to contain at least $\max\{104, \lceil .20M \rceil\}$ transitions. These are descriptive calibration partitions, not individually significant breaks. Whole-prefix scaling is shared by every segment without demeaning; the first target in a segment keeps its preceding observed lag. The constrained AR and GARCH fits have one declared start and no retry. The daily nonregular null and hourly optimization failure retain unavailable primary p values; neither is evidence of stability.

The constant control retains the whole-prefix parameters. The stress arm cycles the fitted regimes with their original transition dwell lengths and a fixed repetition-phase grid. Level, innovation and variance state continue through each switch. The last-to-first wrap is artificial, rather than an observed historical break. Common uniforms are transformed through each arm's own standardized t law; a no-cut source reuses the identical whole-fit path. Both arms retain the forecasting procedures of Section S24. The following complete grids retain failed conditional cells and all F cells.

Table S30: Complete source-by-arm conditional comparison, $W = 52$.

Source	Arm	Full regression Δ [IQR]	Win	Hit	Four-feature Δ [IQR]	Win	Hit
GS-B	C	117.2 [44.3, 191.7]	0.094	0.312	7.2 [−6.6, 20.2]	0.344	0.097
GS-B	S	91.3 [20.0, 174.6]	0.188	0.278	3.2 [−6.6, 50.3]	0.406	0.090
MBS-TOT	C	79.8 [6.1, 214.8]	0.188	0.306	−0.9 [−8.3, 12.5]	0.500	0.097
MBS-TOT	S	140.8 [44.3, 333.9]	0.188	0.285	−0.4 [−6.7, 32.3]	0.531	0.069
CS-TOT	C	124.3 [38.5, 250.2]	0.125	0.340	−3.1 [−12.3, 16.2]	0.594	0.090
CS-TOT	S	124.3 [38.5, 250.2]	0.125	0.340	−3.1 [−12.3, 16.2]	0.594	0.090
FGS-TOT	C	201.0 [128.1, 302.5]	0.062	0.417	1.9 [−11.5, 16.2]	0.469	0.083
FGS-TOT	S	170.1 [27.7, 233.5]	0.062	0.354	−4.4 [−17.6, 12.7]	0.656	0.069
TIPS-L2	C	119.3 [37.1, 213.3]	0.031	0.285	−2.1 [−11.1, 7.6]	0.594	0.069
TIPS-L2	S	101.9 [45.6, 242.1]	0.062	0.319	−0.0 [−5.1, 6.8]	0.500	0.076
TIPS-G11	C	131.3 [62.5, 239.8]	0.125	0.333	1.1 [−4.2, 28.2]	0.406	0.111
TIPS-G11	S	131.3 [62.5, 239.8]	0.125	0.333	1.1 [−4.2, 28.2]	0.406	0.111
TIPS-G2	C	107.9 [47.3, 166.9]	0.125	0.306	−1.5 [−8.2, 9.6]	0.531	0.076
TIPS-G2	S	107.9 [47.3, 166.9]	0.125	0.306	−1.5 [−8.2, 9.6]	0.531	0.076
TIPS-G6L11	C	92.4 [56.4, 168.2]	0.031	0.312	0.1 [−4.3, 8.1]	0.500	0.104
TIPS-G6L11	S	121.4 [57.6, 217.0]	0.062	0.319	3.3 [−3.6, 13.9]	0.406	0.097
GSC-L2	C	116.2 [16.6, 211.1]	0.156	0.285	−0.0 [−5.7, 12.2]	0.500	0.090
GSC-L2	S	123.1 [54.3, 221.6]	0.062	0.306	−2.5 [−7.3, 4.8]	0.656	0.069
GSC-G2L3	C	112.9 [53.6, 164.0]	0.031	0.271	−0.4 [−3.4, 3.3]	0.531	0.090
GSC-G2L3	S	109.1 [52.7, 218.1]	0.062	0.299	10.6 [−8.7, 44.5]	0.438	0.097
GSC-G3L6	C	135.1 [36.7, 237.1]	0.156	0.306	2.8 [−1.3, 14.7]	0.375	0.097
GSC-G3L6	S	166.5 [122.1, 242.0]	0.031	0.368	5.3 [−1.0, 66.4]	0.375	0.104
GSC-G6L7	C	90.8 [30.2, 191.6]	0.156	0.278	1.7 [−3.4, 10.4]	0.438	0.097
GSC-G6L7	S	90.8 [30.2, 191.6]	0.156	0.278	1.7 [−3.4, 10.4]	0.438	0.097
GSC-G7L11	C	92.9 [56.3, 208.5]	0.125	0.243	−4.7 [−9.8, 3.6]	0.688	0.062
GSC-G7L11	S	80.1 [40.1, 196.3]	0.031	0.264	−2.0 [−10.0, 11.9]	0.531	0.083

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Δ: median percent relative loss against the matched F across all 32 pre-specified repetitions, with IQR. Win and QR hit are proportions. FAIL 31/32 is a whole failed cell, with no partial median or win/hit summary; all mandatory repeats must succeed. The failed full regression W260 cells are PDPOSTIPS-G11, repetition 24, in both arms, caused by validation NON-CONVERGENCE. four-feature regression may select a completed candidate under the unchanged pre-specified policy; its failed candidates remain in the records.

Table S31: Complete source-by-arm conditional comparison, $W = 260$.

Source	Arm	Full regression Δ [IQR]	Win	Hit	Four-feature Δ [IQR]	Win	Hit
GS-B	C	$2.9\,[-11.6, 23.1]$	0.438	0.069	$-2.0\,[-16.4, 2.1]$	0.594	0.042
GS-B	S	$4.6\,[-12.9, 57.1]$	0.438	0.104	$0.2\,[-3.6, 5.0]$	0.500	0.062
MBS-TOT	C	$7.0\,[-9.0, 25.6]$	0.406	0.076	$-5.0\,[-15.3, 0.4]$	0.688	0.021
MBS-TOT	S	$18.8\,[2.5, 50.1]$	0.250	0.062	$-0.1\,[-2.6, 15.8]$	0.500	0.035
CS-TOT	C	$2.7\,[-12.5, 39.2]$	0.500	0.125	$-1.2\,[-9.9, 17.8]$	0.594	0.076
CS-TOT	S	$2.7\,[-12.5, 39.2]$	0.500	0.125	$-1.2\,[-9.9, 17.8]$	0.594	0.076
FGS-TOT	C	$13.8\,[-7.5, 45.0]$	0.344	0.104	$-3.0\,[-14.1, 7.0]$	0.688	0.062
FGS-TOT	S	$15.5\,[-17.3, 80.9]$	0.438	0.076	$0.2\,[-3.6, 6.5]$	0.438	0.062
TIPS-L2	C	$9.3\,[-14.2, 24.2]$	0.469	0.104	$-1.7\,[-7.9, 10.6]$	0.594	0.056
TIPS-L2	S	$8.9\,[-5.9, 29.8]$	0.344	0.069	$-0.3\,[-2.0, 2.1]$	0.594	0.049
TIPS-G11	C	**FAIL** 31/32	–	–	$-0.5\,[-9.1, 10.7]$	0.531	0.090
TIPS-G11	S	**FAIL** 31/32	–	–	$-0.5\,[-9.1, 10.7]$	0.531	0.090
TIPS-G2	C	$13.1\,[-11.1, 46.3]$	0.406	0.111	$-0.3\,[-8.9, 4.1]$	0.562	0.062
TIPS-G2	S	$13.1\,[-11.1, 46.3]$	0.406	0.111	$-0.3\,[-8.9, 4.1]$	0.562	0.062
TIPS-G6L11	C	$15.6\,[-3.7, 55.9]$	0.344	0.125	$-0.5\,[-8.8, 13.9]$	0.531	0.069
TIPS-G6L11	S	$20.9\,[-1.9, 62.9]$	0.281	0.132	$1.2\,[-0.4, 10.0]$	0.375	0.056
GSC-L2	C	$10.4\,[-14.0, 42.4]$	0.344	0.069	$-0.8\,[-5.8, 2.7]$	0.625	0.035
GSC-L2	S	$24.5\,[-2.6, 51.8]$	0.312	0.083	$0.8\,[-0.9, 11.2]$	0.406	0.021
GSC-G2L3	C	$1.4\,[-12.9, 13.2]$	0.469	0.118	$-1.4\,[-16.3, 0.5]$	0.688	0.083
GSC-G2L3	S	$-0.6\,[-17.0, 23.3]$	0.500	0.104	$-3.8\,[-9.3, 3.4]$	0.625	0.083
GSC-G3L6	C	$-1.4\,[-17.6, 24.4]$	0.562	0.104	$-1.3\,[-6.3, 0.7]$	0.688	0.056
GSC-G3L6	S	$13.3\,[-7.7, 30.5]$	0.312	0.097	$0.3\,[-0.8, 4.6]$	0.438	0.049
GSC-G6L7	C	$9.1\,[-7.4, 39.0]$	0.438	0.097	$-0.7\,[-3.5, 4.1]$	0.594	0.056
GSC-G6L7	S	$9.1\,[-7.4, 39.0]$	0.438	0.097	$-0.7\,[-3.5, 4.1]$	0.594	0.056
GSC-G7L11	C	$3.6\,[-10.7, 30.2]$	0.438	0.076	$-0.5\,[-5.5, 1.9]$	0.531	0.042
GSC-G7L11	S	$-3.4\,[-19.0, 11.7]$	0.562	0.056	$-5.2\,[-20.8, 3.2]$	0.625	0.062

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Δ: median percent relative loss against the matched F across all 32 pre-specified repetitions, with IQR. Win and QR hit are proportions. FAIL 31/32 is a whole failed cell, with no partial median or win/hit summary; all mandatory repeats must succeed. The failed full regression W260 cells are PDPOSTIPS-G11, repetition 24, in both arms, caused by validation NON-CONVERGENCE. four-feature regression may select a completed candidate under the unchanged pre-specified policy; its failed candidates remain in the records.

Table S32: Corrected-rank F benchmark: all 52 source-by-arm-history cells.

Source	Arm	History	Complete / required	Loss [IQR]	Hit [IQR]
GS-B	C	52	32/32	$0.09015\,[0.07863, 0.1076]$	$0.05556\,[0.02778, 0.09722]$
GS-B	C	260	32/32	$0.07989\,[0.06798, 0.09176]$	$0.04167\,[0, 0.09722]$
GS-B	S	52	32/32	$0.08146\,[0.06128, 0.1055]$	$0.05556\,[0.02778, 0.08333]$
GS-B	S	260	32/32	$0.08697\,[0.0705, 0.1085]$	$0.04167\,[0, 0.1528]$
MBS-TOT	C	52	32/32	$0.05759\,[0.0481, 0.07073]$	$0.05556\,[0, 0.08333]$
MBS-TOT	C	260	32/32	$0.05332\,[0.04593, 0.05875]$	$0.02083\,[0, 0.05903]$
MBS-TOT	S	52	32/32	$0.0464\,[0.04173, 0.06648]$	$0.03472\,[0, 0.08333]$
MBS-TOT	S	260	32/32	$0.04574\,[0.04135, 0.05452]$	$0.01389\,[0, 0.05556]$
CS-TOT	C	52	32/32	$0.08467\,[0.07035, 0.1165]$	$0.04167\,[0, 0.09722]$
CS-TOT	C	260	32/32	$0.08322\,[0.06532, 0.1016]$	$0.04861\,[0, 0.08681]$
CS-TOT	S	52	32/32	$0.08467\,[0.07035, 0.1165]$	$0.04167\,[0, 0.09722]$
CS-TOT	S	260	32/32	$0.08322\,[0.06532, 0.1016]$	$0.04861\,[0, 0.08681]$
FGS-TOT	C	52	32/32	$0.05269\,[0.04771, 0.06046]$	$0.04861\,[0.01389, 0.06944]$
FGS-TOT	C	260	32/32	$0.04924\,[0.0459, 0.05809]$	$0.04167\,[0.01389, 0.06944]$
FGS-TOT	S	52	32/32	$0.06601\,[0.05181, 0.07969]$	$0.02778\,[0.01389, 0.07292]$
FGS-TOT	S	260	32/32	$0.05653\,[0.05491, 0.07435]$	$0.04167\,[0, 0.1111]$
TIPS-L2	C	52	32/32	$0.09575\,[0.07975, 0.1155]$	$0.04167\,[0.02778, 0.07292]$
TIPS-L2	C	260	32/32	$0.0859\,[0.07622, 0.09682]$	$0.04167\,[0.02431, 0.08333]$
TIPS-L2	S	52	32/32	$0.08608\,[0.0697, 0.111]$	$0.05556\,[0.02778, 0.08333]$
TIPS-L2	S	260	32/32	$0.07837\,[0.06901, 0.1028]$	$0.04167\,[0.01389, 0.08681]$
TIPS-G11	C	52	32/32	$0.09022\,[0.06905, 0.1083]$	$0.05556\,[0.01389, 0.08333]$
TIPS-G11	C	260	32/32	$0.08016\,[0.07022, 0.1084]$	$0.05556\,[0.02778, 0.1111]$
TIPS-G11	S	52	32/32	$0.09022\,[0.06905, 0.1083]$	$0.05556\,[0.01389, 0.08333]$
TIPS-G11	S	260	32/32	$0.08016\,[0.07022, 0.1084]$	$0.05556\,[0.02778, 0.1111]$
TIPS-G2	C	52	32/32	$0.1007\,[0.07528, 0.121]$	$0.05556\,[0.01389, 0.08681]$
TIPS-G2	C	260	32/32	$0.08534\,[0.06626, 0.1093]$	$0.05556\,[0.02431, 0.08681]$

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Source	Arm	History	Complete / required	Loss [IQR]	Hit [IQR]
TIPS-G2	S	52	32/32	0.1007 [0.07528, 0.121]	0.05556 [0.01389, 0.08681]
TIPS-G2	S	260	32/32	0.08534 [0.06626, 0.1093]	0.05556 [0.02431, 0.08681]
TIPS-G6L11	C	52	32/32	0.09185 [0.08051, 0.1171]	0.07639 [0.05208, 0.09722]
TIPS-G6L11	C	260	32/32	0.0826 [0.07391, 0.09724]	0.0625 [0.02778, 0.1007]
TIPS-G6L11	S	52	32/32	0.09133 [0.07505, 0.1247]	0.06944 [0.03819, 0.09722]
TIPS-G6L11	S	260	32/32	0.08318 [0.07664, 0.1058]	0.04167 [0, 0.08333]
GSC-L2	C	52	32/32	0.04554 [0.04079, 0.05505]	0.04861 [0.02431, 0.08681]
GSC-L2	C	260	32/32	0.04131 [0.0388, 0.04593]	0.02778 [0, 0.08333]
GSC-L2	S	52	32/32	0.04493 [0.0366, 0.04898]	0.04861 [0.01389, 0.08333]
GSC-L2	S	260	32/32	0.03859 [0.03435, 0.04521]	0.02778 [0, 0.06944]
GSC-G2L3	C	52	32/32	0.06574 [0.05149, 0.0898]	0.05556 [0.02778, 0.08333]
GSC-G2L3	C	260	32/32	0.0645 [0.05369, 0.08905]	0.06944 [0.01389, 0.1111]
GSC-G2L3	S	52	32/32	0.06149 [0.05254, 0.08228]	0.04861 [0.01389, 0.08681]
GSC-G2L3	S	260	32/32	0.0616 [0.05365, 0.0779]	0.04861 [0, 0.08333]
GSC-G3L6	C	52	32/32	0.05913 [0.05199, 0.08991]	0.05556 [0.01389, 0.1111]
GSC-G3L6	C	260	32/32	0.05448 [0.04917, 0.06724]	0.04167 [0.02431, 0.07292]
GSC-G3L6	S	52	32/32	0.05589 [0.04396, 0.07093]	0.07639 [0.02778, 0.1111]
GSC-G3L6	S	260	32/32	0.04793 [0.045, 0.05747]	0.04861 [0, 0.09722]
GSC-G6L7	C	52	32/32	0.05322 [0.04505, 0.06819]	0.04167 [0.01389, 0.07292]
GSC-G6L7	C	260	32/32	0.0482 [0.04246, 0.05957]	0.04861 [0.02431, 0.06944]
GSC-G6L7	S	52	32/32	0.05322 [0.04505, 0.06819]	0.04167 [0.01389, 0.07292]
GSC-G6L7	S	260	32/32	0.0482 [0.04246, 0.05957]	0.04861 [0.02431, 0.06944]
GSC-G7L11	C	52	32/32	0.05278 [0.04627, 0.06312]	0.02778 [0.01389, 0.06944]
GSC-G7L11	C	260	32/32	0.04761 [0.04315, 0.05204]	0.02778 [0, 0.05903]
GSC-G7L11	S	52	32/32	0.05369 [0.04537, 0.06031]	0.04167 [0, 0.08333]
GSC-G7L11	S	260	32/32	0.04759 [0.04422, 0.05826]	0.04861 [0.01042, 0.1007]

C: fitted constant control; S: parameter-switching stress. Source codes omit PDPOS. Loss is the median across repetitions of the within-repetition mean pinball score, with IQR; Hit is the corresponding breach-rate median and IQR. All 52 cells have 32/32 valid final F receipts. Loss units follow each source-specific whole-prefix SD normalization; different sources are not pooled. No failed-cell partial median is substituted.

Loss summaries are medians across repetitions within each source, then equal-weight medians across every required source. They are not the controlled grid's ratio-of-means estimand. A required repetition failure makes its whole cell unavailable; successful subsets do not replace missing full-grid results. The raw weekly screening threshold is met, while multiplicity adjustment retains fewer rejections. The near-parity constant-control prediction is unsupported. The two-history switching prediction and the daily/hourly comparison are unassessable. Constant controls themselves generate large losses, so instability is not necessary for losses in these procedures. The empirical low-frequency mechanism remains unidentified.

S29 Paired menu isolation: full source inventory

S29.1 Isolating the penalty menu

The weekly failures motivate a closer look at the penalty menu in calibrated simulations. Earlier stationary comparisons already produce large short-history regression losses. The fitted constant and switching arms use thirteen pre-evaluation weekly calibrations. The earlier stability screen rejects seven series pointwise and two after Holm correction. Supplement Sections S27–S28 retain the earlier calibrated simulations and stability diagnostics (Tables S27–S32).

The paired menu experiment changes only the ridge candidate menu and the stated exact-tie rule. It adds penalties one, ten and the intercept-only limit to full regression. Both regressions use the six-point menu. All paths, innovations, source calibrations and forecasting clocks match the earlier experiment. Thirty-two paired repetitions contribute to each complete source median. This isolates the executed menu comparison within these calibrated laws. The expanded procedure can choose ordinary I. Stronger finite penalties and access to I are not separately identified; near-parity loss alone supplies no evidence of added feature value.

Table S33: Penalty-menu isolation on matched calibrated paths. Entries are the median relative loss (%) across all 13 source medians, with the descriptive source IQR in brackets.

Arm	History	Regression	Original menu Median [IQR]	Expanded menu Median [IQR]	Missing sources Original / expanded
Constant	52	Full regression	116.24 [92.90, 124.34]	0.21 [-0.52, 0.98]	0 / 0
Constant	52	Four-feature	-0.02 [-1.55, 1.73]	0.44 [-1.69, 1.90]	0 / 0
Constant	260	Full regression	NA	NA	1 / 1
Constant	260	Four-feature	-1.21 [-1.69, -0.54]	-1.13 [-1.69, -0.64]	0 / 0
Switching	52	Full regression	121.40 [101.92, 131.26]	-0.23 [-1.88, 0.98]	0 / 0
Switching	52	Four-feature	-0.03 [-2.03, 3.22]	0.38 [-2.17, 3.17]	0 / 0
Switching	260	Full regression	NA	NA	1 / 1
Switching	260	Four-feature	-0.27 [-0.66, 0.23]	-0.28 [-0.75, 0.48]	0 / 0

Each source median uses all 32 matched repetitions with relative loss $100(\overline{L}_{QR}/\overline{L}_F - 1)$; positive values mean higher regression loss, and NA retains unavailable all-source summaries without substituting successful subsets. Displays are rounded, while pre-specified decisions use saved full-precision operands; the two regression variants retain their stated predictor sets. The expanded four-feature exact-tie rule prefers stronger penalties, unlike its original rule; saved exact ties: 0, changed selections: 0.

The expanded full-regression medians move close to historical headroom in both arms (Table S33). Figure S1 displays these across-source medians and their descriptive spread. Both arms improve in every source. The constant-arm near-parity prediction and the paired-improvement prediction hold. The switching-minus-constant difference misses the pre-specified five-percentage-point increase. This outcome supports the menu explanation for these simulated losses; it leaves the empirical frequency mechanism unresolved.

At history 260, one required full-regression repetition fails in each arm under the old low penalty. Both summaries across all thirteen sources remain unavailable. The experiment retains every failed cell and performs no retry. Earlier daily moment-qualification and hourly optimization failures also remain unavailable. Absence of a five-point switching increment is not evidence of equivalence.

The identification qualification above concerns the paired simulated menu intervention. The additional financial finite-menu diagnostic is described in Section S36.

Procedure and complete-source inventory. Both QR variants use penalties $\{.001, .01, .1, 1, 10, \infty\}$; the infinite candidate is ordinary intercept-only I, distinct from corrected-rank F. Full regression requires every candidate to complete. Four-feature regression selects from its completed eligible candidates, as before. Exact ties now prefer stronger shrinkage in both variants; no actual exact tie occurred. The switching-phase denominator remains thirty-two. Two pilot repetitions selected the full thirty-two formal repetitions before launch. The simulated feature clock is inherited hourly geometry despite weekly calibration parameters; it is not a literal weekly-calendar replay.

The design preceded every new path calculation. M1 requires the constant-arm history-52 full-regression median within five percent of F. M2 requires the switching median at least five percentage points higher. M3 requires both aggregate improvements and at least twelve jointly improving sources. M1 and M3 hold; M2 fails. Full history-260 regression summaries remain unavailable after the required TIPS-G11 repetition fails in both arms. Successful repetitions or twelve-source subsets never replace them.

Constant=C; switching=S. Positive percentages mean higher loss than F. NA retains the whole failed cell; source medians and old results remain separate.
Table S34 retains the complete source, arm, history and method inventory.



Figure S1: Paired penalty-menu isolation on unchanged calibrated paths. The expanded menu removes the large short-history full-regression loss in both arms. Longer-history full-regression summaries remain unavailable because a required repetition fails in each arm, with no successful-subset replacement.

Table S34: Every menu-isolation source/arm/history/method cell. Relative-loss medians use all required repetitions.

Source	Arm	History	Predictors	Original median	Expanded median	Complete/required	State
GS-B	C	52	33	117.16	4.24	32/32	COMPLETE
GS-B	C	52	4	7.20	8.70	32/32	COMPLETE
GS-B	C	260	33	2.90	0.10	32/32	COMPLETE
GS-B	C	260	4	-2.02	-1.92	32/32	COMPLETE
GS-B	S	52	33	91.35	2.47	32/32	COMPLETE
GS-B	S	52	4	3.22	3.17	32/32	COMPLETE
GS-B	S	260	33	4.56	0.56	32/32	COMPLETE
GS-B	S	260	4	0.23	0.09	32/32	COMPLETE
MBS-TOT	C	52	33	79.84	-0.49	32/32	COMPLETE
MBS-TOT	C	52	4	-0.87	-0.70	32/32	COMPLETE
MBS-TOT	C	260	33	7.02	0.97	32/32	COMPLETE
MBS-TOT	C	260	4	-5.00	-5.15	32/32	COMPLETE
MBS-TOT	S	52	33	140.83	-0.59	32/32	COMPLETE
MBS-TOT	S	52	4	-0.44	-0.92	32/32	COMPLETE
MBS-TOT	S	260	33	18.77	0.91	32/32	COMPLETE
MBS-TOT	S	260	4	-0.06	-0.22	32/32	COMPLETE
CS-TOT	C	52	33	124.34	-3.01	32/32	COMPLETE
CS-TOT	C	52	4	-3.07	-3.07	32/32	COMPLETE
CS-TOT	C	260	33	2.68	-0.19	32/32	COMPLETE
CS-TOT	C	260	4	-1.21	-1.13	32/32	COMPLETE
CS-TOT	S	52	33	124.34	-3.01	32/32	COMPLETE
CS-TOT	S	52	4	-3.07	-3.07	32/32	COMPLETE
CS-TOT	S	260	33	2.68	-0.19	32/32	COMPLETE
CS-TOT	S	260	4	-1.21	-1.13	32/32	COMPLETE
FGS-TOT	C	52	33	201.01	4.44	32/32	COMPLETE
FGS-TOT	C	52	4	1.90	1.90	32/32	COMPLETE
FGS-TOT	C	260	33	13.76	1.34	32/32	COMPLETE
FGS-TOT	C	260	4	-2.98	-2.98	32/32	COMPLETE
FGS-TOT	S	52	33	170.12	-0.23	32/32	COMPLETE
FGS-TOT	S	52	4	-4.36	-4.36	32/32	COMPLETE
FGS-TOT	S	260	33	15.48	-0.40	32/32	COMPLETE
FGS-TOT	S	260	4	0.23	0.48	32/32	COMPLETE
TIPS-L2	C	52	33	119.29	-0.54	32/32	COMPLETE
TIPS-L2	C	52	4	-2.10	-2.10	32/32	COMPLETE
TIPS-L2	C	260	33	9.32	0.64	32/32	COMPLETE
TIPS-L2	C	260	4	-1.69	-1.69	32/32	COMPLETE
TIPS-L2	S	52	33	101.92	0.53	32/32	COMPLETE
TIPS-L2	S	52	4	-0.03	0.38	32/32	COMPLETE
TIPS-L2	S	260	33	8.88	-0.38	32/32	COMPLETE
TIPS-L2	S	260	4	-0.27	-0.28	32/32	COMPLETE
TIPS-G11	C	52	33	131.26	0.98	32/32	COMPLETE
TIPS-G11	C	52	4	1.11	1.14	32/32	COMPLETE
TIPS-G11	C	260	33	NA	NA	31/32	FAILED
TIPS-G11	C	260	4	-0.48	-0.49	32/32	COMPLETE
TIPS-G11	S	52	33	131.26	0.98	32/32	COMPLETE
TIPS-G11	S	52	4	1.11	1.14	32/32	COMPLETE
TIPS-G11	S	260	33	NA	NA	31/32	FAILED
TIPS-G11	S	260	4	-0.48	-0.49	32/32	COMPLETE
TIPS-G2	C	52	33	107.85	-4.00	32/32	COMPLETE
TIPS-G2	C	52	4	-1.55	-1.69	32/32	COMPLETE
TIPS-G2	C	260	33	13.09	0.12	32/32	COMPLETE
TIPS-G2	C	260	4	-0.29	-0.43	32/32	COMPLETE
TIPS-G2	S	52	33	107.85	-4.00	32/32	COMPLETE
TIPS-G2	S	52	4	-1.55	-1.69	32/32	COMPLETE
TIPS-G2	S	260	33	13.09	0.12	32/32	COMPLETE
TIPS-G2	S	260	4	-0.29	-0.43	32/32	COMPLETE
TIPS-G6L11	C	52	33	92.36	1.00	32/32	COMPLETE
TIPS-G6L11	C	52	4	0.11	0.44	32/32	COMPLETE
TIPS-G6L11	C	260	33	15.63	2.53	32/32	COMPLETE
TIPS-G6L11	C	260	4	-0.54	-0.64	32/32	COMPLETE
TIPS-G6L11	S	52	33	121.40	3.18	32/32	COMPLETE
TIPS-G6L11	S	52	4	3.27	3.68	32/32	COMPLETE
TIPS-G6L11	S	260	33	20.88	1.90	32/32	COMPLETE
TIPS-G6L11	S	260	4	1.23	1.23	32/32	COMPLETE
GSC-L2	C	52	33	116.24	0.21	32/32	COMPLETE
GSC-L2	C	52	4	-0.02	1.85	32/32	COMPLETE
GSC-L2	C	260	33	10.40	-0.21	32/32	COMPLETE
GSC-L2	C	260	4	-0.80	-1.22	32/32	COMPLETE
GSC-L2	S	52	33	123.05	-1.88	32/32	COMPLETE
GSC-L2	S	52	4	-2.47	-2.17	32/32	COMPLETE
GSC-L2	S	260	33	24.51	0.08	32/32	COMPLETE
GSC-L2	S	260	4	0.81	0.87	32/32	COMPLETE
GSC-G2L3	C	52	33	112.85	0.74	32/32	COMPLETE
GSC-G2L3	C	52	4	-0.43	-0.28	32/32	COMPLETE

Source	Arm	History	Predictors	Original median	Expanded median	Complete/required	State
GSC-G2L3	C	260	33	1.42	0.36	32/32	COMPLETE
GSC-G2L3	C	260	4	-1.39	-1.02	32/32	COMPLETE
GSC-G2L3	S	52	33	109.05	0.02	32/32	COMPLETE
GSC-G2L3	S	52	4	10.61	10.73	32/32	COMPLETE
GSC-G2L3	S	260	33	-0.63	-1.22	32/32	COMPLETE
GSC-G2L3	S	260	4	-3.84	-3.95	32/32	COMPLETE
GSC-G3L6	C	52	33	135.13	0.90	32/32	COMPLETE
GSC-G3L6	C	52	4	2.78	2.43	32/32	COMPLETE
GSC-G3L6	C	260	33	-1.44	-1.16	32/32	COMPLETE
GSC-G3L6	C	260	4	-1.26	-1.37	32/32	COMPLETE
GSC-G3L6	S	52	33	166.48	4.12	32/32	COMPLETE
GSC-G3L6	S	52	4	5.26	6.42	32/32	COMPLETE
GSC-G3L6	S	260	33	13.34	1.34	32/32	COMPLETE
GSC-G3L6	S	260	4	0.31	0.61	32/32	COMPLETE
GSC-G6L7	C	52	33	90.84	-0.32	32/32	COMPLETE
GSC-G6L7	C	52	4	1.73	2.14	32/32	COMPLETE
GSC-G6L7	C	260	33	9.12	0.02	32/32	COMPLETE
GSC-G6L7	C	260	4	-0.66	-0.75	32/32	COMPLETE
GSC-G6L7	S	52	33	90.84	-0.32	32/32	COMPLETE
GSC-G6L7	S	52	4	1.73	2.14	32/32	COMPLETE
GSC-G6L7	S	260	33	9.12	0.02	32/32	COMPLETE
GSC-G6L7	S	260	4	-0.66	-0.75	32/32	COMPLETE
GSC-G7L11	C	52	33	92.90	-0.52	32/32	COMPLETE
GSC-G7L11	C	52	4	-4.71	-4.35	32/32	COMPLETE
GSC-G7L11	C	260	33	3.61	-0.36	32/32	COMPLETE
GSC-G7L11	C	260	4	-0.51	-0.55	32/32	COMPLETE
GSC-G7L11	S	52	33	80.11	-2.78	32/32	COMPLETE
GSC-G7L11	S	52	4	-2.03	-2.67	32/32	COMPLETE
GSC-G7L11	S	260	33	-3.41	0.20	32/32	COMPLETE
GSC-G7L11	S	260	4	-5.15	-5.15	32/32	COMPLETE

S30 CFTC sources, eligibility and information dates

The source is the CFTC TFF futures-only historical backcast for 2006–2016 plus annual 2017–2026 files. The retained observations begin in 2010. All eleven original source records remain public; no other new financial data were acquired.

A complete generated calendar contains 877 Monday-labeled report weeks, from 2010-01-04 through 2026-10-19. Economic position dates are retained separately. A code needs one valid observation per week, a stable exchange and stable contract units. Missing weeks, duplicates and changes disqualify a contract rather than trigger interpolation or splicing. Known consolidation prefixes 13874,20974,12460 are excluded before ranking. This conservative rule is broader than the documented plus-suffixed codes. Full-period mean open interest ranks survivors, with literal-code ties. Seventeen of 141 observed codes qualify; the top-twenty cap never changes that count. The complete code-level eligibility and exclusions preserve every screening decision.

Validation indices are 553–604; evaluation indices are 611–870. All labels use horizon six, leaving six unissued origins between the segments. Validation position dates are 2020-08-11–2021-08-03; the last validation label ends 2021-09-14. Evaluation position dates are 2021-09-21–2026-09-08; the last evaluation label ends 2026-10-20. Training labels at refit r span r-H-W+1 through r-H. Monday labels index the inherited feature calendar; they do not claim Monday public availability. Actual positions are normally measured Tuesday and released Friday at 15:30 Eastern. Holiday, shutdown and ION-related publication delays prevent a first-release interpretation.

The selected category series are listed below. Net positions equal dealer long minus short in raw contract counts. Equal-weight median percentage changes preserve within-contract units; they are not common-dollar portfolio risk.

Code	Latest contract name	Mean open interest	Weeks
044601	UST 5Y NOTE - CHICAGO BOARD OF TRADE	3479174.2	877
043602	UST 10Y NOTE - CHICAGO BOARD OF TRADE	3344673.4	877
042601	UST 2Y NOTE - CHICAGO BOARD OF TRADE	2170647.9	877
045601	FED FUNDS - CHICAGO BOARD OF TRADE	1282757.7	877
020601	UST BOND - CHICAGO BOARD OF TRADE	999944.0	877
099741	EURO FX - CHICAGO MERCANTILE EXCHANGE	503616.5	877
1170E1	VIX FUTURES - CBOE FUTURES EXCHANGE	356897.7	877
097741	JAPANESE YEN - CHICAGO MERCANTILE EXCHANGE	210863.1	877
096742	BRITISH POUND - CHICAGO MERCANTILE EXCHANGE	197853.5	877
095741	MEXICAN PESO - CHICAGO MERCANTILE EXCHANGE	174995.0	877
090741	CANADIAN DOLLAR - CHICAGO MERCANTILE EXCHANGE	166229.3	877
232741	AUSTRALIAN DOLLAR - CHICAGO MERCANTILE EXCHANGE	154802.4	877
33874A	E-MINI S&P 400 STOCK INDEX - CHICAGO MERCANTILE EXCHANGE	75156.1	877
240743	NIKKEI STOCK AVERAGE YEN DENOM - CHICAGO MERCANTILE EXCHANGE	59766.6	877
092741	SWISS FRANC - CHICAGO MERCANTILE EXCHANGE	57197.3	877
098662	USD INDEX - ICE FUTURES U.S.	52473.9	877
112741	NZ DOLLAR - CHICAGO MERCANTILE EXCHANGE	44292.0	877

The six methods are corrected-rank F, ordinary I, four- and thirty-three-predictor regression, and finite delayed calibration of historical headroom and full regression. Both regressions use the six-point menu. The paired ACI wrappers retain the existing finite calibration with delayed matured feedback; no asymptotic coverage claim is attached. The complete 204 method cells finish without a numerical retry. The full cell CSV and trace ZIP retain absolute scores, breaches, selected penalties and gamma values.

For the CFTC delayed wrappers, gamma belongs to $\{.001, .005, .01, .05\}$. Completed candidates rank by validation coverage-distance, then pinball loss, then smaller gamma. Initialization requires at least

nineteen matured scores. Updates wait for the full six-record label, and evaluation carries the validation state. The six unissued gap origins create no artificial reports or feedback events. These settings accompany histories 52/260, 52 validation origins, 260 evaluation origins and 13-record refits; none is retuned after evaluation.

Two narrow official-format clarifications follow the original design and acquisition but precede every forecast. The first accepts only the official midnight suffix after the date; the second accepts integer counts written with an all-zero decimal suffix. Neither changes a mathematical count, sample threshold, screen or forecast setting. Original failed technical preparation receipts remain visible. This is not a claim of fully observation-blind ingestion design.

S31 Complete model confidence sets

The pre-specified ninety-percent MCS uses the range statistic and 1,000 stationary-bootstrap draws, with block length $\max\{H, \lceil n^{1/3} \rceil\}$ in arch 8.0.0. Deterministic source/grid seeds and method order are recorded in the design. Every claimed complete report uses its entire declared origin grid and common target. Missing stored losses are derived only from saved reports, without issuing forecasts or fitting models. Every one of 98 comparisons completes. The 60 known unavailable slots remain unavailable, not statistical exclusions.

Distinct procedure labels retain their classification even when a selected penalty is infinite or reports coincide. Exact duplicate loss columns enter the library once; all associated procedure labels then receive the same p-value and inclusion partition. Conditional slots use their declared procedure identity. Matched F/I membership, literal uniqueness and unique-loss-class membership differ. Across the entire inventory, 90/98 sets contain matched F/I, 2/98 have a sole literal F/I member, and 4/98 have a sole loss class containing F/I. Conditional exclusion is 197/683 complete slots. Related sources and overlapping tasks provide no independent cross-grid test or simultaneous future guarantee.

Tables S36–S45 retain every source-specific membership result and method p-value, including unavailable procedures. **Complete pre-specified MCS membership and p-values.** All 98 source/period/H/W comparisons are retained. Bold p-values denote included procedures using the saved literal rule $p > 0.10$; ordinary type denotes exclusion. Values are rounded to three decimals only for display. Membership uses unrounded saved values; exact p-values are public in `results/v10/mcs/report-values.json`. NA denotes a known unavailable procedure, never a statistical exclusion; — means that procedure was not pre-specified in that particular grid. Multi-member exact-loss classes carry a superscript equal to the representative method ID on *every* member, including the representative. Unmarked values are singleton classes. Method IDs are display aliases only; all full labels and classifications appear in the final legend. The pre-specified procedure uses 1000 stationary-bootstrap replications, confidence 0.90, statistic R, and block length $\max\{H, \lceil n^{1/3} \rceil\}$. Analytic identical-loss or single-method cases retain their saved status; no new inference is performed for this table.

FX original. Each method column is shown once for every row of this group.

Table S36: FX original; method block 1/3. Full-grid $n = 6625$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. SAV history	Cal. SAV regression	SAV headroom	Expanded full reg.	Endpoint plug-in	Four-feature Legacy multi-fold	Finite ridge	Expanded ridge	One-SE ridge
fx_hourly	6	720	0.000	0.002^{M02}	0.000	0.019	0.000	0.040^{M10}	0.000	0.010	NA	0.040^{M10}	0.040^{M10}	0.002^{M02}
fx_hourly	6	2880	0.000	0.000^{M02}	0.000	**0.385**	0.000	**1.000**^{M10}	0.000	0.001	—	**1.000**^{M10}	**1.000**^{M10}	0.000^{M02}
fx_hourly	24	720	**0.696**	**0.760**	0.009	0.000	**0.792**	0.009^{M10}	0.011	**0.250**	NA	0.009^{M10}	0.009^{M10}	0.009^{M10}
fx_hourly	24	2880	**0.360**	**0.360**	**0.360**	**0.360**	**0.360**	**0.360**^{M10}	0.005	**0.360**	—	**0.360**^{M10}	**0.360**^{M10}	0.078^{M16}

Table S37: FX original; method block 2/3. Full-grid $n = 6625$.

Source	Horizon	History	Multi-fold one-SE	Filtered paths	Peak-tail fit	Quantile forest	Nearest states	Original full reg.	Scaled neighbors	Fixed peak	Fixed transl.	State headroom	State peak	State transl.
fx_hourly	6	720	—	NA	NA	0.002	0.014	0.040^{M10}	0.016	0.000	0.004	0.002	0.000	0.014
fx_hourly	6	2880	**1.000**^{M10}	0.000	0.000	0.001	0.054	**1.000**^{M10}	0.039	0.000	0.000	—	—	—
fx_hourly	24	720	—	NA	NA	**0.199**	**0.445**	0.009^{M10}	**0.347**	0.000	**0.696**	**0.792**	0.067	**0.792**
fx_hourly	24	2880	0.078^{M16}	0.079	0.002	**0.153**	0.078	**0.360**^{M10}	0.052	0.000	**0.360**	—	—	—

Table S38: FX original; method block 3/3. Full-grid $n = 6625$.

Source	Horizon	History	Rolling headroom	Rolling peak	Rolling transl.	Mean combination	Median combination
fx_hourly	6	720	0.002	0.000	0.003	**1.000**	0.047
fx_hourly	6	2880	0.000	0.000	0.000	0.039	0.057
fx_hourly	24	720	**0.792**	0.000	**1.000**	**0.696**	**0.551**
fx_hourly	24	2880	**0.360**	0.000	**0.360**	**1.000**	**0.360**

FX early 2015. Each method column is shown once for every row of this group.

Table S39: FX early 2015; method block 1/1. Full-grid $n = 1440$.

Source	Horizon	History	Hist. headroom	Intercept only	Legacy multi-fold	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE	Original full reg.	Original roll. hist.
fx_hourly	6	720	0.063	**0.675**	NA	**0.675**$^{\mathrm{M14}}$	**0.675**$^{\mathrm{M14}}$	**0.628**	—	—	**1.000**
fx_hourly	6	2880	0.000	0.000	—	**1.000**$^{\mathrm{M14}}$	**1.000**$^{\mathrm{M14}}$	0.009	**1.000**$^{\mathrm{M14}}$	**1.000**$^{\mathrm{M14}}$	—
fx_hourly	24	720	0.014	**0.242**	NA	**1.000**$^{\mathrm{M14}}$	**1.000**$^{\mathrm{M14}}$	**1.000**$^{\mathrm{M14}}$	—	—	—
fx_hourly	24	2880	0.000	**0.910**	—	**1.000**$^{\mathrm{M14}}$	**1.000**$^{\mathrm{M14}}$	**0.991**$^{\mathrm{M16}}$	**0.991**$^{\mathrm{M16}}$	—	—

Treasury bills. Each method column is shown once for every row of this group.

Table S40: Treasury bills; method block 1/3. Full-grid $n = 277$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. SAV history	Cal. SAV regression	SAV headroom	Expanded full reg.	Endpoint plug-in	Four-feature	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE
nyfed_weekly	6	52	**0.809**^{M01}	**0.400**^{M02}	0.000	0.004	0.045	**0.940**^{M10}	**0.810**	**0.770**	**0.940**^{M10}	**0.400**^{M02}	**0.400**^{M02}	**0.400**^{M02}
nyfed_weekly	6	260	**0.663**	0.073	**0.144**	**0.194**	**0.273**	**0.452**^{M10}	**0.982**	**0.622**	**0.452**^{M10}	**0.452**^{M10}	**0.622**^{M16}	**0.622**^{M16}
nyfed_weekly	24	52	**0.862**^{M01}	**0.626**^{M02}	**0.926**	0.004	**0.253**	**0.925**^{M10}	**0.211**	**0.150**	**0.925**^{M10}	**0.626**^{M02}	**0.626**^{M02}	**0.626**^{M02}
nyfed_weekly	24	260	0.066	**0.165**^{M02}	NA	NA	**0.261**	**0.186**^{M10}	**0.186**	**0.178**	**0.186**^{M10}	**0.186**^{M10}	**0.165**^{M02}	**0.165**^{M02}

Table S41: Treasury bills; method block 2/3. Full-grid $n = 277$.

Source	Horizon	History	Filtered paths	Peak-tail fit	Quantile forest	Nearest states	Original full reg.	Scaled neighbors	Fixed peak	Fixed transl.	State headroom	State peak	State transl.	Rolling headroom
nyfed_weekly	6	52	**0.940**	NA	**0.940**	**0.809**^{M01}	0.012	0.016	**0.743**	**0.767**	**0.863**	**0.940**	**0.802**	**0.597**
nyfed_weekly	6	260	**0.897**	**0.982**	**0.622**	**0.845**	0.028	**0.157**	**0.982**	**0.687**	**0.957**	**0.982**	**0.913**	**0.633**
nyfed_weekly	24	52	**0.150**	NA	**0.977**	**0.862**^{M01}	**0.211**	0.025	**0.925**	**0.979**	**0.862**	**0.926**	**0.977**	**0.925**
nyfed_weekly	24	260	0.000	0.000	0.066	**0.261**	**0.201**	0.000	**0.487**	0.072	**1.000**	**0.336**	**0.487**	0.066

Table S42: Treasury bills; method block 3/3. Full-grid $n = 277$.

Source	Horizon	History	Rolling peak	Rolling transl.	Mean combination	Median combination
nyfed_weekly	6	52	**1.000**	**0.516**	**0.940**	**0.809**^{M01}
nyfed_weekly	6	260	**1.000**	**0.663**	**0.622**	**0.848**
nyfed_weekly	24	52	**0.926**	**1.000**	**0.925**	**0.862**^{M01}
nyfed_weekly	24	260	**0.624**	0.066	**0.336**	**0.321**

Other NYFed panels. Each method column is shown once for every row of this group.

Table S43: Other NYFed panels; method block 1/1. Full-grid $n = 277$.

Source	Horizon	History	Hist. headroom	Intercept only	Finite ridge	Expanded ridge	One-SE ridge	Multi-fold one-SE	Expanded full reg.	Four-feature	Endpoint plug-in	Original full reg.	Mean combination	Median combination
PDPOSCS-TOT	6	52	**0.312**	**1.000**^{M02}	**0.312**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.312**^{M14}	**0.312**	**0.172**	0.095	**0.312**	NA
PDPOSCS-TOT	6	260	**0.582**	**0.780**^{M02}	**0.243**^{M14}	**0.243**^{M14}	**0.780**^{M02}	**0.780**^{M02}	**0.243**^{M14}	**0.780**	0.000	**0.697**	**1.000**	NA
PDPOSCS-TOT	24	52	**0.898**	**0.898**^{M02}	**0.780**^{M14}	**0.898**^{M02}	**0.898**^{M02}	**0.898**^{M02}	**0.780**^{M14}	**0.898**	**1.000**	0.085	**0.780**	NA
PDPOSCS-TOT	24	260	0.000	**1.000**^{M02}	0.010^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	0.010^{M14}	0.000	0.000	0.003	0.000	NA
PDPOSFGS-TOT	6	52	**0.248**	**1.000**^{M02}	**0.868**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.868**^{M14}	**0.968**	**0.974**	0.014	**0.492**	NA
PDPOSFGS-TOT	6	260	0.017	0.039	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	0.001	0.000	**0.709**	**0.709**	NA
PDPOSFGS-TOT	24	52	**0.521**	**0.390**^{M02}	**0.307**^{M14}	**0.307**^{M14}	**0.390**^{M02}	**0.390**^{M02}	**0.307**^{M14}	**0.521**	**1.000**	**0.133**	**0.390**	NA
PDPOSFGS-TOT	24	260	0.010	**0.772**	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	**0.772**^{M14}	0.000	**1.000**	**0.470**	**0.425**	NA
PDPOSGSC-G2L3	6	52	**1.000**	**0.208**	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	**0.189**^{M14}	0.019	0.035	0.018	**0.189**	NA
PDPOSGSC-G2L3	6	260	**0.676**	**0.676**	**0.118**^{M14}	**0.118**^{M14}	**0.676**^{M16}	**0.676**^{M16}	**0.118**^{M14}	**0.348**	**1.000**	**0.118**^{M14}	**0.521**	NA
PDPOSGSC-G2L3	24	52	**1.000**	**0.280**^{M02}	0.004^{M14}	0.004^{M14}	0.069	**0.280**^{M02}	0.004^{M14}	0.005	0.065	0.004^{M14}	0.069	NA
PDPOSGSC-G2L3	24	260	**0.204**	**0.204**^{M02}	**0.271**^{M14}	**0.204**^{M02}	**0.204**^{M02}	**0.204**^{M02}	**0.271**^{M14}	**0.204**	**1.000**	**0.245**	**0.271**	NA
PDPOSGSC-G3L6	6	52	**0.624**	**0.921**	**0.426**^{M14}	**0.426**^{M14}	**0.602**^{M16}	**0.602**^{M16}	**0.426**^{M14}	0.033	0.078	0.025	**1.000**	NA
PDPOSGSC-G3L6	6	260	**0.746**	**0.437**^{M02}	**0.710**^{M14}	**0.710**^{M14}	**0.437**^{M02}	**0.437**^{M02}	**0.710**^{M14}	**0.437**	**0.461**	**0.710**^{M14}	**1.000**	NA
PDPOSGSC-G3L6	24	52	**0.471**	**0.471**^{M02}	**0.460**^{M14}	**0.460**^{M14}	**0.460**^{M14}	**0.471**^{M02}	**0.460**^{M14}	0.039	**1.000**	0.009	0.039	NA
PDPOSGSC-G3L6	24	260	**0.103**	**0.127**	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**1.000**^{M14}	**0.103**	**0.997**	**0.678**	**0.703**	NA
PDPOSGSC-G6L7	6	52	**0.962**	**1.000**^{M02}	**0.119**^{M14}	**0.119**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**0.119**^{M14}	**0.119**	**0.962**	0.000	**0.258**	NA
PDPOSGSC-G6L7	6	260	**0.366**	**0.366**	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089^{M14}	0.089	**1.000**	0.089^{M14}	**0.364**	NA
PDPOSGSC-G6L7	24	52	**1.000**	**0.305**	0.000^{M14}	0.000^{M14}	0.000^{M14}	**0.356**	0.000^{M14}	0.000	**0.929**	0.000^{M14}	**0.356**	NA
PDPOSGSC-G6L7	24	260	**1.000**	**0.407**	**0.131**^{M14}	**0.131**^{M14}	**0.162**^{M16}	**0.162**^{M16}	**0.131**^{M14}	**0.192**	**0.407**	**0.131**^{M14}	**0.407**	NA
PDPOSGSC-G7L11	6	52	**1.000**	**0.166**^{M02}	**0.143**^{M14}	**0.143**^{M14}	**0.143**^{M14}	**0.166**^{M02}	**0.143**^{M14}	**0.166**	**0.807**	0.014	**0.166**	NA
PDPOSGSC-G7L11	6	260	**0.806**	**0.702**^{M02}	**0.609**^{M14}	**0.609**^{M14}	**0.702**^{M02}	**0.702**^{M02}	**0.609**^{M14}	**0.806**	**0.806**	**0.609**^{M14}	**1.000**	NA
PDPOSGSC-G7L11	24	52	**0.459**	**0.136**^{M02}	**0.459**^{M14}	**0.136**^{M02}	**0.136**^{M02}	**0.136**^{M02}	**0.459**^{M14}	0.092	**1.000**	**0.120**	**0.371**	NA
PDPOSGSC-G7L11	24	260	**1.000**	**0.488**	0.073^{M14}	0.073^{M14}	**0.253**^{M16}	**0.253**^{M16}	0.073^{M14}	0.078	**0.795**	0.073^{M14}	**0.253**	NA
PDPOSGSC-L2	6	52	**0.305**	**0.614**^{M02}	0.000^{M14}	0.000^{M14}	**0.614**^{M02}	**0.614**^{M02}	0.000^{M14}	**1.000**	0.002	0.000^{M14}	**0.130**	NA
PDPOSGSC-L2	6	260	**0.862**	**0.830**^{M02}	**0.862**^{M14}	**0.862**^{M14}	**0.830**^{M02}	**0.830**^{M02}	**0.862**^{M14}	**0.200**	**1.000**	**0.193**	**0.486**	NA
PDPOSGSC-L2	24	52	**1.000**	**0.202**^{M02}	**0.202**^{M14}	**0.202**^{M14}	**0.202**^{M14}	**0.202**^{M02}	**0.202**^{M14}	**0.209**	**0.928**	0.002	**0.174**	NA
PDPOSGSC-L2	24	260	**0.359**	**0.359**^{M02}	**0.359**^{M14}	**0.359**^{M02}	**0.359**^{M02}	**0.359**^{M02}	**0.359**^{M14}	**0.359**	**1.000**	**0.149**	**0.359**	NA
PDPOSMBS-TOT	6	52	**1.000**	**0.576**^{M02}	**0.464**^{M14}	**0.576**^{M02}	**0.576**^{M02}	**0.576**^{M02}	**0.464**^{M14}	**0.576**	**0.576**	0.000	**0.576**	NA
PDPOSMBS-TOT	6	260	**0.902**	**0.415**^{M02}	**0.192**^{M14}	**0.192**^{M14}	**0.415**^{M02}	**0.415**^{M02}	**0.192**^{M14}	**0.493**	**0.798**	**0.192**^{M14}	**1.000**	NA
PDPOSMBS-TOT	24	52	**0.110**	**0.110**^{M02}	0.072^{M14}	0.072^{M14}	0.072^{M14}	**0.110**^{M02}	0.072^{M14}	**0.110**	**1.000**	0.018	**0.110**	NA
PDPOSMBS-TOT	24	260	**0.784**	**0.784**^{M02}	**0.784**^{M14}	**0.784**^{M02}	**0.784**^{M02}	**0.784**^{M02}	**0.784**^{M14}	**0.784**	**1.000**	**0.784**	**0.784**	NA
PDPOSTIPS-G11	6	52	**0.832**	**1.000**	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**^{M14}	**0.282**	**0.261**	0.000	**0.238**	NA
PDPOSTIPS-G11	6	260	0.011	**1.000**^{M02}	**0.106**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**1.000**^{M02}	**0.106**^{M14}	0.096	0.014	0.068	**0.135**	NA
PDPOSTIPS-G11	24	52	**0.673**	**0.673**^{M02}	0.037^{M14}	0.037^{M14}	**0.673**^{M02}	**0.673**^{M02}	0.037^{M14}	0.018	**1.000**	0.004	0.037	NA
PDPOSTIPS-G11	24	260	**0.449**	**1.000**	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.449**^{M14}	**0.506**	0.012	**0.449**^{M14}	**0.506**	NA
PDPOSTIPS-G2	6	52	**1.000**	**0.482**^{M02}	**0.241**^{M14}	**0.241**^{M14}	**0.482**	**0.482**^{M02}	**0.241**^{M14}	**0.445**	**0.245**	0.004	**0.245**	NA
PDPOSTIPS-G2	6	260	**0.701**	**1.000**	0.057^{M14}	0.057^{M14}	0.057^{M14}	0.057^{M14}	0.057^{M14}	**0.581**	**0.215**	0.057^{M14}	0.001	NA
PDPOSTIPS-G2	24	52	**0.283**	**1.000**^{M02}	**0.169**^{M14}	**0.169**^{M14}	**1.000**^{M02}	**1.000**^{M02}	**0.169**^{M14}	**0.426**	**0.255**	0.006	**0.169**	NA
PDPOSTIPS-G2	24	260	**1.000**	**0.663**^{M02}	0.078^{M14}	0.078^{M14}	**0.663**^{M02}	**0.663**^{M02}	0.078^{M14}	**0.200**	**0.701**	0.078^{M14}	**0.419**	NA
PDPOSTIPS-G6L11	6	52	**0.437**	0.000^{M02}	**0.437**^{M14}	0.000^{M02}	0.000^{M02}	0.000^{M02}	**0.437**^{M14}	**0.437**	**0.162**	0.025	**1.000**	NA
PDPOSTIPS-G6L11	6	260	0.018	**0.101**^{M02}	**0.101**^{M14}	**0.101**^{M02}	**0.101**^{M02}	**0.101**^{M02}	**0.101**^{M14}	**0.101**	0.000	**0.887**	**1.000**	NA
PDPOSTIPS-G6L11	24	52	**0.280**	0.001^{M02}	**0.231**^{M14}	**0.231**^{M14}	0.001^{M02}	0.001^{M02}	**0.231**^{M14}	**0.231**	**0.477**	0.071	**1.000**	NA
PDPOSTIPS-G6L11	24	260	0.000	0.000^{M02}	0.000^{M14}	0.000^{M02}	0.000^{M02}	0.000^{M02}	0.000^{M14}	0.000	0.000	**1.000**	0.000	NA
PDPOSTIPS-L2	6	52	**0.921**	**0.921**^{M02}	**0.921**^{M14}	**0.921**^{M02}	**0.921**^{M02}	**0.921**^{M02}	**0.921**^{M14}	**0.258**	**1.000**	0.000	0.075	NA
PDPOSTIPS-L2	6	260	**0.460**	**0.294**	0.047^{M14}	0.047^{M14}	**0.301**^{M16}	**0.301**^{M16}	0.047^{M14}	**0.460**	**1.000**	0.047^{M14}	**0.460**	NA
PDPOSTIPS-L2	24	52	**0.247**	**0.247**	0.015^{M14}	0.015^{M14}	0.015^{M14}	**0.247**	0.015^{M14}	**0.230**	**1.000**	0.000	0.015	NA
PDPOSTIPS-L2	24	260	**0.427**	**0.248**^{M02}	0.000^{M14}	0.000^{M14}	**0.248**^{M02}	**0.248**^{M02}	0.000^{M14}	0.012	**1.000**	0.000^{M14}	0.002	NA

Same-record daily FX. Each method column is shown once for every row of this group.

Table S44: Same-record daily FX; method block 1/1. Full-grid $n = 501$.

Source	Horizon	History	Hist. headroom	Full regression	Four-feature
fx_daily	6	52	**1.000**	0.005	0.005
fx_daily	6	260	**0.604**	0.087	**1.000**
fx_daily	24	52	**1.000**	0.011	0.041
fx_daily	24	260	**1.000**	**0.347**	**0.581**

CFTC. Each method column is shown once for every row of this group.

Table S45: CFTC; method block 1/1. Full-grid $n = 260$.

Source	Horizon	History	Hist. headroom	Intercept only	Cal. history	Cal. full regression	Full regression	Four-feature
cftc:020601	6	52	**0.774**	**0.309**	**0.133**	**1.000**	0.053	**0.133**
cftc:020601	6	260	**0.242**	**0.234**	**1.000**	**0.819**	**0.242**	**0.242**
cftc:042601	6	52	**1.000**	**0.540**	**0.264**	**0.430**	**0.264**	**0.540**
cftc:042601	6	260	**0.957**	**1.000**	**0.957**	**0.892**	**0.629**	0.068
cftc:043602	6	52	**1.000**	**0.258**^{M08}	**0.101**	**0.101**	**0.258**^{M08}	**0.126**
cftc:043602	6	260	**0.773**	**0.171**	**0.927**	**1.000**	**0.200**	**0.221**
cftc:044601	6	52	**1.000**	**0.323**	**0.323**	**0.196**	**0.196**	0.043
cftc:044601	6	260	**1.000**	**0.457**	**0.457**	0.092	0.069	**0.229**
cftc:045601	6	52	**0.810**	**1.000**^{M08}	**0.120**	**0.810**	**1.000**^{M08}	**0.810**
cftc:045601	6	260	**1.000**	**0.164**^{M08}	**0.967**	**0.967**	**0.164**^{M08}	**0.164**^{M08}
cftc:090741	6	52	**1.000**	**0.578**^{M08}	**0.578**	**0.578**	**0.578**^{M08}	**0.160**
cftc:090741	6	260	**0.915**	**0.372**	**0.915**	**1.000**	**0.962**	**0.985**
cftc:092741	6	52	**0.986**	**0.986**^{M08}	**1.000**	**0.978**	**0.986**^{M08}	**0.986**^{M08}
cftc:092741	6	260	**0.761**	**0.761**	**0.761**	**0.761**	**0.724**	**1.000**
cftc:095741	6	52	**1.000**	0.067	0.067	0.048	0.067	**0.142**
cftc:095741	6	260	**0.823**	**1.000**	**0.542**	0.016	**0.121**	**0.811**
cftc:096742	6	52	**1.000**	**0.876**	0.009	0.009	**0.876**	**0.876**
cftc:096742	6	260	**0.982**	**0.324**	**0.162**	**1.000**	0.059	**0.966**
cftc:097741	6	52	**0.790**	**1.000**	**0.263**	**0.368**	**0.263**	**0.368**
cftc:097741	6	260	**0.955**	**0.848**	**0.955**	**0.955**	**0.612**	**1.000**
cftc:098662	6	52	**0.986**	**1.000**^{M09}	0.013	0.009	**0.986**	**1.000**^{M09}
cftc:098662	6	260	0.016	0.020^{M08}	0.011	0.011	0.020^{M08}	**1.000**
cftc:099741	6	52	**0.624**	**0.624**	0.065	**1.000**	**0.510**	**0.624**
cftc:099741	6	260	**1.000**	**0.861**^{M08}	0.095	0.095	**0.861**^{M08}	**0.861**^{M08}
cftc:112741	6	52	**1.000**	**0.178**	**0.131**	**0.131**	**0.178**	**0.178**
cftc:112741	6	260	**1.000**	**0.298**	**0.831**	**0.634**	0.037	**0.884**
cftc:1170E1	6	52	**0.981**	**0.582**	**1.000**	**0.582**	**0.182**	**0.582**
cftc:1170E1	6	260	0.000	0.000	**0.110**	**0.179**	**0.278**	**1.000**
cftc:232741	6	52	**1.000**	**0.959**	0.053	0.003	0.099	0.066
cftc:232741	6	260	**0.976**	**0.732**	**0.976**	**1.000**	**0.732**	**0.976**
cftc:240743	6	52	0.042	0.014	0.042	0.001	0.005	**1.000**
cftc:240743	6	260	0.000	**1.000**^{M08}	0.019	0.036	**1.000**^{M08}	**1.000**^{M08}
cftc:33874A	6	52	**1.000**	**0.241**	**0.241**	**0.177**	**0.241**	0.014
cftc:33874A	6	260	0.003	**0.926**	**0.926**	**0.921**	**1.000**	**0.926**

Complete method legend.

F/I are canonical matched aliases within each comparison only; this does not equate forecasts across sources. Class codes: H=matched historical F/I; R=other historical (rolling F); C=conditional procedure; B=other benchmark; U=unclassified known NA. Conditional procedures retain class C even when they select infinity or share exact losses with F/I. Superscripts identify exact full-loss classes, not semantic method mergers. Known NA IDs retain their original full spelling.

Short ID	Procedure and source context	Class
M01	Corrected historical headroom	H
M02	Ordinary empirical headroom	H
M03	Delayed calibration of historical headroom	C
M04	Delayed calibration of conditional regression	C
M05	Headroom quantile recursion	C
M06	CFTC: delayed calibration of historical headroom	C
M07	CFTC: delayed calibration of full regression	C
M08	CFTC: full regression	C
M09	CFTC: four-feature regression	C
M10	Expanded-menu full regression	C
M11	Endpoint plug-in	B
M12	Four-feature regression	C
M13	Earlier-period rolling validation rule	C
M14	Earlier-period minimum-loss rule, original menu	C
M15	Earlier-period minimum-loss rule, intercept limit added	C
M16	Earlier-period one-standard-error rule	C
M17	Earlier-period rolling one-standard-error rule	C
M18	Earlier-period original three-penalty regression	C
M19	Earlier-period original historical reference	R
M20	Daily FX: full regression	C
M21	Daily FX: four-feature regression	C
M22	Filtered historical paths	C
M23	Volatility-scaled peak tail model	B
M24	Quantile forest	C
M25	Original nearest-state prediction	C
M26	Original conditional regression	C
M27	Increment-scaled nearest-state prediction	C
M28	Original fixed historical peak quantile	B
M29	Original fixed translated paths	B
M30	Original one-state historical headroom	C
M31	Original one-state historical peak quantile	C
M32	Original one-state translated paths	C
M33	Original rolling historical headroom	R
M34	Original rolling historical peak quantile	B
M35	Original rolling translated paths	B
M36	Panel: expanded-menu full regression	C
M37	Panel: four-feature regression	C
M38	Panel: endpoint plug-in	B
M39	Panel: original full regression	C
M40	Mean forecast combination	C
M41	Median forecast combination	C

The display is exhaustive but descriptive across related, overlapping tasks; it does not pool independent-source p-values or establish future risk coverage. The machine display map preserves full-precision entries, NA metadata, every class representative and per-comparison storage aliases.

S32 Fixed limit costs and full parameter regions

For each comparison, let B be the issue-known floor, Y the realized sampled peak, C_m a saved method report, and $m_F = \overline{C_F - B} > 0$ the matched F report's mean reserved headroom. All four quantities use the same source-native position units (currency or contract counts). The normalized per-issue cost is

$$J_m(\kappa, \beta, c) = \kappa \frac{\overline{C_m - B}}{m_F} + \beta\, \overline{\mathbf{1}\{Y > C_m\}} + c \frac{\overline{(Y - C_m)_+}}{m_F}, \qquad b = \beta m_F.$$

Reserve, strict breach frequency and excess enter separately. The fixed breach charge b is in those same position-value units; each β scales it to that comparison's evaluation-sample F headroom. Charges therefore differ across comparisons. Costs apply per issued report; overlapping horizons are not added as separate cash flows. The fixed grid is $\kappa \in \{.01, .025, .05, .1, .2, .5, 1\}$, $\beta \in \{0, .01, .05, .1, .25, .5, 1, 2, 5\}$ and $c \in \{.5, 1, 2\}$. At $\kappa = .05, \beta = 0, c = 1$, the cost difference from F is the pinball-loss difference divided by m_F. Other weights need not preserve that ranking. The panels compare fixed reports and do not estimate a cost-optimal policy.

All 98 comparisons have a positive scale and all 976 complete methods enter 189 grid points, giving 184,464 cost rows. Sixteen missing-floor inputs are recovered only from their pre-specified, fingerprinted same-origin F copies. Their complete origin, target and floor arrays match exactly. Other missing or mismatched floors would make the whole comparison unavailable. No missing scale uses an epsilon.

The public `results/v10/cost/cost-grid.csv.zip` contains every complete method's normalized mean cost, its F reference and strict cheaper/equal status at all coefficients. `primary-regions.csv` retains all 18,522 planned primary grid rows, including status fields. Primitive reserve amounts, strict breach counts/rates and excess amounts appear in `primitive-stats.json`. Source-group figures use the full planned comparison denominators. The primary comparison is fixed expanded-menu full regression for hourly/weekly/CFTC sources and the original saved regression for daily FX. Other methods retain separate regions and are never replaced by a minimum-cost oracle.

S33 Retained accuracy and hit diagnostics

These are the unchanged earlier tests and Murphy plots moved from the main article. The p-values retain their original named adjustment families; no MCS adjustment is retroactively assigned to them.

Table S47: Accuracy and hit diagnostics for explicitly named reports. Panel A reports paired working-reference tests; Panel B uses the fixed nonoverlap grid for hit tests. A better average score does not imply fewer breaches.

Panel A: paired contrast	DM p	HLN p	Named adjustment / p
FX original main comparison	2.20e-07	2.31e-07	Two-main comparison Bonferroni: 4.39e-07
FX matched history	4.47e-09	4.83e-09	Original matched Holm family: 2.86e-07
Treasury original main comparison	0.00761	0.00939	Original matched Holm family: 0.320

Original matched Holm uses 72 contrasts. The later regrouped matched family is separate; earlier-period writing-stage HAC comparisons have neither a new HLN field nor a new family adjustment.

Panel B: report	Grid hits	Grid issues	Kupiec p	Christoffersen CC p
FX regression	50	947	0.695	0.636
FX original rolling reference	49	947	0.807	0.905
FX matched fixed reference	49	947	0.807	0.646
Treasury regression	7	40	0.00413	0.0136
Treasury historical reference	4	40	0.199	0.262

Hit tests use the full fixed nonoverlap grid; the conditional-coverage likelihood conditions on its initial hit. Equality to the nominal tail rate is the stated diagnostic null, distinct from the population upper-quantile inequality. The grid does not create independence.

S34 Reproduction, availability and retained extended results

Reproduction and run records are retained verbatim in the v10b reproduction records.

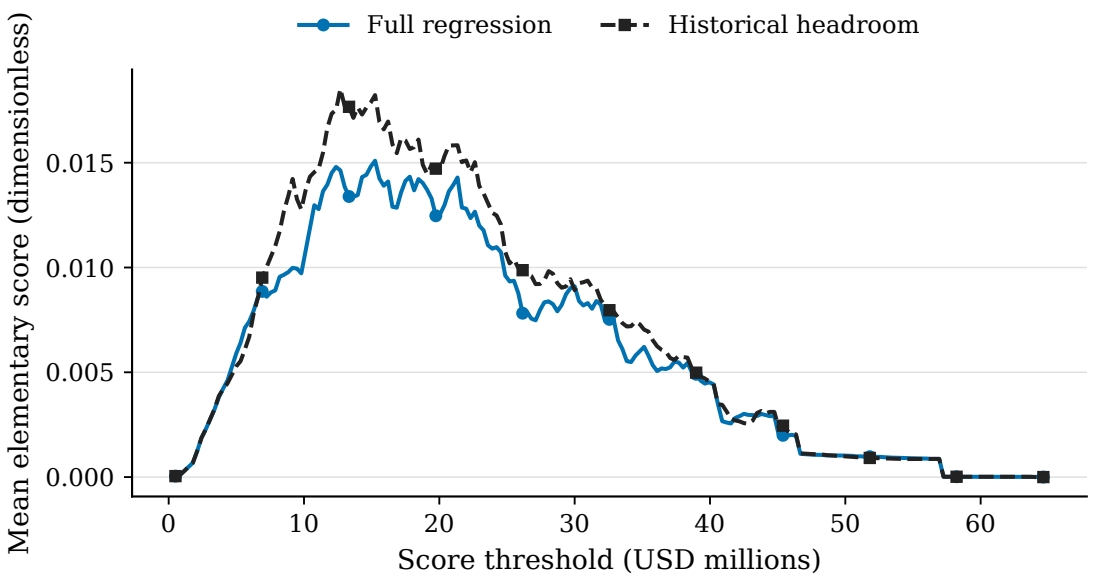


Figure S2: Murphy diagram for the main short-horizon FX comparison. The full-breakpoint calculation finds crossings; the graph is a display of saved scores. Average pinball superiority does not imply dominance across elementary scores.

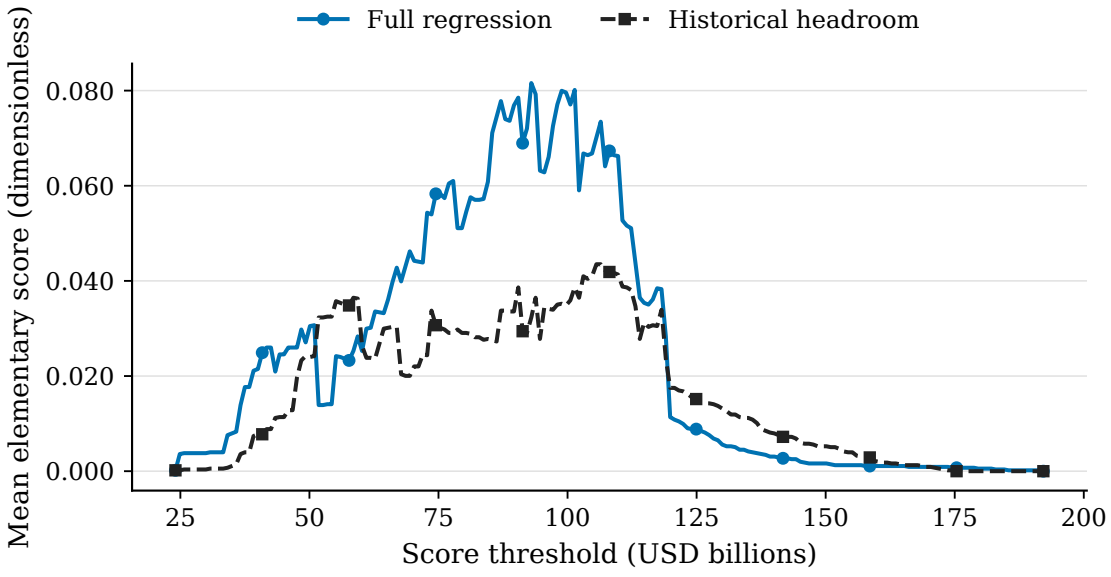


Figure S3: Murphy diagram for the main Treasury-bill comparison. The saved full-breakpoint comparison also crosses. A single average-score ordering leaves other consistent-score preferences unresolved.

Editable sources, aggregate results, original designs and code accompany the paper; favorable, adverse and failed outputs remain accessible.

Document regeneration checks saved aggregates without raw observations; it does not establish recomputed forecast accuracy or independent replication.

Code is available at the current repository entry; the root reproduction instructions identify required raw inputs and executable checks.

Full financial replay requires compatible inputs. A separately authorized replay requires the original FX member and Treasury snapshot to match their recorded fingerprints and definitions. Restricted FX observations and private per-origin forecasts or fitted traces are not redistributed. Downloaded economic dates do not establish verified first-release availability; a matching series name does not establish snapshot identity.

Routes to complete evidence. The v6 replay guide, original conditional records, supervisor rule records and extended-results index identify the complete retained tables, failure inventories and longer technical discussions. Individual-dealer external validation and prospective first-release validation remain unmet.

S35 Additional saved evidence on history length

The earlier confirmation exercise applies its fixed pooled criteria to heavy-tailed, regime-switching and compound-Poisson innovation laws. Table S48 reports every prewritten criterion, including the failed universal ordering prediction. Its history regions use method-specific thresholds estimated in the earlier source study; they are not an oracle rule for the financial data.

Table S48: Saved confirmation criteria for excess loss relative to matched historical headroom under additional synthetic innovation laws. Counts use the original fixed pooled denominators, including all counterexamples.

Prewritten criterion	Successes	Missing	Required	Result
Positive excess loss in the low-history region	47/48	0	80%	Supported
Negative excess loss in the high-history region	21/24	0	60%	Supported
Four-feature loss never exceeds full-regression loss	42/48	0	100%	Not supported

History regions use the method-specific thresholds fixed in the linked original design, conditional on the earlier threshold exercise; the history index excludes predictor dimension. Full and four-feature regressions retain different original penalty menus, so their ordering does not isolate predictor dimension. The universal ordering prediction is not supported; the complete records retain all counterexamples and candidate failures.

The unchanged source records retain the complete synthetic results, counterexamples and candidate failures. The original confirmation design records the fixed laws, history regions, penalty menus and decision rules. The local design commits precede their numerical runs; the later public file uploads are not public preregistration.

The complete FX history contrasts retain both horizons and every tested window. Both longer-horizon methods lose their gain again at the longest history. The four-feature short-horizon first cutpoint is left-censored and not testable, so the original confirmation across all curves is not established. These comparisons use the inspected period and provide neither evidence from an additional dealer nor a fresh financial holdout.

S36 Retained scope details and additional empirical checks

Earlier scope, retained verbatim. The following paragraphs reproduce the earlier manuscript's scope discussion verbatim. They describe the evidence available before the additional empirical checks summarized below; statements about untested work retain that historical meaning.

The FX evidence describes a single market maker's client-flow record. Its original evaluation period was viewed before the conditional methods were specified. Rules for the earlier 2015 comparison followed the original results. The full-prefix backcast uses penalties selected after its issue dates and contains the earlier backcast. Its target paths were covered by previous scoring. Both exercises remain retrospective.

The New York Fed weekly categories share a reporting system and economic components; they are related aggregate tasks, not independent dealer replications. Treasury comparisons replay the downloaded economic-date vintage. Historical publication-time information is not reconstructed. Aggregation can offset gross dealer exposure; sampled net inventory also differs from settlement cash needs. Valuation and reporting-composition changes are not separated from trading. Sampled peaks omit intervening excursions. Daily last records also differ from verified closes or intraday maxima; the source timezone and release vintage are unknown.

Daily aggregation jointly changes physical histories, horizons, update cadence, calendar predictors, evaluation span and sampled targets. The same-record comparison leaves a pure frequency effect unidentified.

The design gives no causal interpretation or individual-predictor effects. The financial procedures provide no coverage guarantee under their dependence and calibration scheme. Trading and funding gains are not measured. Scoring weights are evaluation preferences, not estimated intervention or funding costs. Independent dealer records and prospective observations would broaden validation; finer exposure and cost data would connect reports to inventory actions.

Short overlapping validation samples and research-stage selection limit tuning stability and confirmatory inference. Longer weekly outer validation windows remain untested. The simulation designs use different menus, clocks, calibration laws and summary statistics. The paired menu experiment isolates that procedure change within the calibrated laws; empirical causation remains unresolved. The new parameter-switching process repeats fitted dwell lengths and adds artificial last-to-first transitions. Its stability screen uses a plug-in model-based approximation. Moment qualification and successful optimization leave the assumed law and boundary-estimate uncertainty unresolved. Failed cells and unavailable diagnostic comparisons retain those statuses.

The statement about untested longer weekly validation windows records the earlier scope only. The additional CFTC validation-length comparison now tests longer windows on the unchanged evaluation period, with its complete outcomes and failures linked below. It does not establish prospective tuning stability.

CFTC categories provide an independent reporting system, not independent individual dealers. Full-history continuity creates a retrospective survivor universe. Open-interest ranking only orders the eligible contracts here. Conservative consolidation-family exclusions leave seventeen eligible contracts. Raw contracts have different exposure units; normalized comparisons do not recover common-dollar risks. Downloaded position dates omit actual historical first-release vintages and reporting delays. Two ingestion-schema clarifications followed acquisition but preceded every forecast and model-loss result; their original failed technical attempts remain documented.

Coefficients are fixed preferences; no estimated funding return, optimal intervention policy or trading-profit claim follows. The normalization uses evaluation-sample F headroom; ex ante institutional charges remain untested. MCS inference requires suitable loss dependence. The stationary bootstrap, modest samples and research-stage specification choices limit its nominal interpretation. Membership and cross-task fractions do not certify future coverage or simultaneous best-model status.

Issued-report consistency. Separately fitted horizon forecasts can violate the ordering of true peak quantiles. Short-history FX regression produces horizon crossings on 640 evaluation origins. Label nesting, floor coverage and strict-breach definitions provide additional checks on the issued reports. The complete horizon-coherence comparisons remain in Section S22; autoregressive replay and empirical-rank details remain in Sections S7 and S8.

Additional empirical checks and complete results. Complete source-cell records, candidate means, prewritten decisions and execution receipts are retained in the loss-decomposition results, the validation-length results and the tail-level results. The decomposition evaluates every declared penalty on the same evaluation origins. Its best finite-penalty evaluation score is an optimistic retrospective diagnostic, not a usable selection rule. The infinite candidate is ordinary empirical headroom and is excluded from that finite oracle; the selection component may therefore be negative. Longer validation changes selection rather than evaluation dates or refit schedules. Tail-level checks change empirical ranks, fitted targets and scores together; the declared sample-maximum cap supplies no coverage guarantee. These new empirical checks require all validation candidates to complete, including for four-feature regression; the earlier simulated policy above is unchanged. Required numerical failures remain failed whole procedure cells, without retries or successful-subset substitution. Failed fits leave the highest-level CFTC threshold and two-period FX criterion unresolved; completed lower-level outputs do not replace them.

S37 Model confidence sets: retained main comparison

Joint comparison leaves matched historical methods alongside alternatives in most sets, with a single historical method surviving in only a small minority. These results support retaining historical controls while leaving many alternatives uneliminated under the fixed bootstrap approximation. Non-elimination does not establish equal predictive risk. Table S49 reports surviving methods, with identical observed-loss patterns explained separately in Supplement Section S31. Supplement Section S31 retains every set, method p-value, equivalence class and prior unavailable method.

These confidence sets use the original comparison inventory; the later validation rules are reported separately in Section S38.

Table S49: Ninety-percent model confidence sets across all planned comparisons. Counts retain their comparison or conditional-procedure denominators.

Group	Complete / planned	Matched history included	Sole historical method	Conditional excluded	Unavailable procedures
FX original	4/4	2/4 (50.0%)	0/4 (0.0%)	39/66 (59.1%)	6
FX earlier period	4/4	3/4 (75.0%)	0/4 (0.0%)	1/15 (6.7%)	2
Treasury bills	4/4	4/4 (100.0%)	0/4 (0.0%)	11/74 (14.9%)	4
Other NYFed panels	48/48	46/48 (95.8%)	0/48 (0.0%)	110/384 (28.6%)	48
Same-record daily FX	4/4	4/4 (100.0%)	2/4 (50.0%)	5/8 (62.5%)	0
CFTC	34/34	31/34 (91.2%)	0/34 (0.0%)	31/136 (22.8%)	0
All comparisons	98/98	90/98 (91.8%)	2/98 (2.0%)	197/683 (28.8%)	60

Matched history includes corrected and ordinary empirical headroom reports, excluding rolling references and nearest-neighbor baselines; conditional exclusions count complete procedures. Identical losses can leave several methods tied, so the sole-method column counts only sets containing exactly one historical procedure. Supplement Section S31 retains all tied loss patterns, secondary membership counts and unavailable procedures; related comparisons are not independent replications.

S38 Validation extensions and a retrospective decision rule

Post-failure solver check. The solver change designed after the failure was not adopted. Only 6 of 10 previously converged fits reproduced forecasts within 10^{-6} pointwise relative error. Residual and duality-gap thresholds were not relaxed. The pre-specified gate therefore prevented all three failed 99% comparisons from running with this change. Their original unavailable values remain, leaving both aggregate tail-level conditions undetermined. The complete numerical check retains every agreement failure and unrun comparison.

Longer validation in the New York Fed series. Across-series median loss falls under longer validation at both horizons, extending the direction found in CFTC. Table S50 reports both regression variants, including the failed short-horizon median-improvement condition for the four-feature variant. Figure S4 retains every sequence beside the CFTC comparisons rather than pooling their evidence.

Table S50: New York Fed validation-length comparisons on the unchanged evaluation sample. Entries are across-series median excess loss relative to corrected historical headroom, with changes in percentage points and selections of the ordinary empirical limit. Every method/horizon row contains all thirteen series.

Method	H	52 weeks (%)	156 weeks (%)	Change (pp)	∞: 52/156	Median screen	Shrinkage screen
Full regression	6	+6.68	-1.13	-7.81	6 / 9	Pass	Pass
Full regression	24	+40.47	+5.02	-35.45	3 / 9	Pass	Pass
Four-feature	6	+3.26	-0.95	-4.22	5 / 7	Fail	Pass
Four-feature	24	+24.88	+6.21	-18.67	4 / 9	Pass	Pass

The two screens are the pre-specified across-series median-improvement and increased infinite-penalty-selection conditions. Arm-median changes are not medians of paired series changes.

The original validation origins are 329–380; longer validation uses 243–398 at the short horizon and 225–380 at the longer horizon. Evaluation remains 405–681, with 277 issues. Thus the short-horizon endpoint also advances eighteen weeks and changes the validation refit phase; this is not a causal length-only comparison. The longer-horizon endpoint is unchanged. Differences between arm medians also need not describe a typical series: for full regression, the paired-change median is +0.00 and -44.10 percentage points, respectively. All source, candidate and failure records remain in the complete validation-length results; related weekly categories are not independent dealer replications.

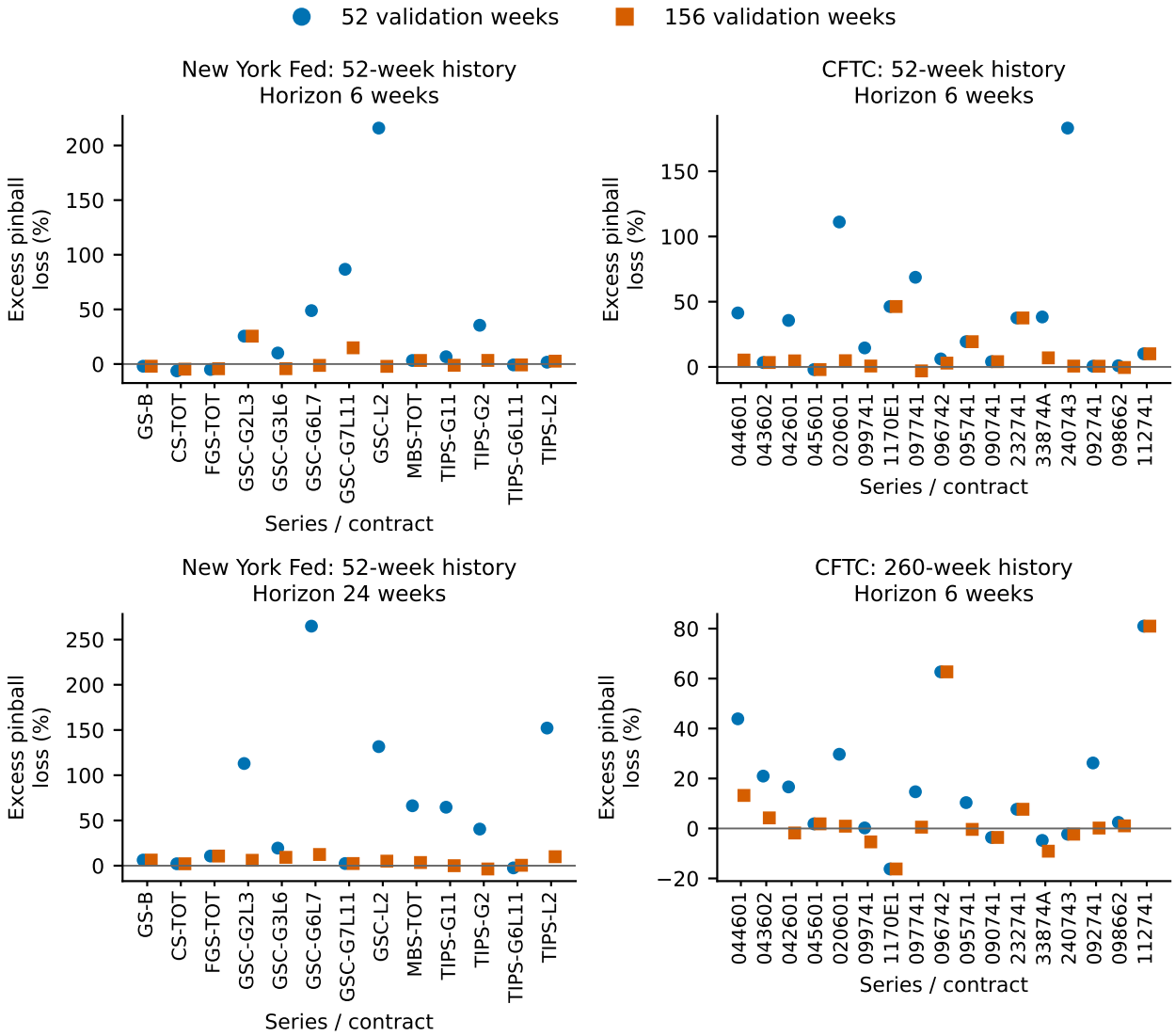


Figure S4: Full-regression losses under short and long validation samples, with each reporting system retaining its own sources and histories. Points use unchanged evaluation origins; positive excess means higher loss than matched corrected historical headroom. The New York Fed short-horizon extension also changes the endpoint and refit phase; shortened source labels omit the common PDPOS prefix, so GS-B means PDPOSGS-B.

The retrospective validation protocol. The rule uses the six-point penalty menu, longer validation, and a historical default. Weekly validation contains 156 issues and FX validation contains 5,760; regression replaces corrected historical headroom only when its validation advantage exceeds one paired HAC standard error. Selection and the gate share validation data, and the decision stays fixed over the evaluation segment. Infinite shrinkage gives ordinary empirical headroom, so it must pass the same gate to replace the corrected-rank reference. These settings combine existing forecasting and validation methods without a nominal protection guarantee.

All 62 planned comparisons complete, including 60 weekly cells and both FX periods. The weekly median moves from +14.63% for ordinary regression with the six-point menu to 0.00% for the protocol. However, 8 of 60 weekly cells exceed the historical loss by more than 10%, and the worst excess is +80.98%. The median condition therefore passes while the tail condition fails. The protocol returns to history in 32 weekly cells and uses regression in 28, as Table S51 shows. The adverse cells comprise 7 CFTC comparisons and 1 New York Fed comparison. Both FX periods retain their ordinary-regression gains, 14.77% and 16.62%, meeting the gain-retention condition.

Table S51: Historical fallback and adverse outcomes by source, horizon and training history. Counts use every planned comparison in each row; the last column counts protocol excess loss above ten percent.

Source / period	H	W	Historical fallback	Median excess (%)	Adverse count
Treasury bills	6	52	1 / 1	+0.00	0
Other NY Fed series	6	52	8 / 12	+0.00	1
Treasury bills	24	52	1 / 1	+0.00	0
Other NY Fed series	24	52	10 / 12	+0.00	0
CFTC	6	52	6 / 17	+0.00	4
CFTC	6	260	6 / 17	+0.00	3
FX, original	6	2880	0 / 1	-14.77	0
FX, earlier	6	2880	0 / 1	-16.62	0

The adverse CFTC cells are 1170E1, 095741, 232741 and 112741 at the shorter history, and 044601, 096742 and 112741 at the longer history. The remaining cell is the New York Fed two-to-three-year Treasury coupon category at the short horizon. The complete protocol results retain every comparison, selected penalty, paired gate and fallback route. The protocol was designed after inspecting earlier results, including these evaluation periods, so this is a retrospective test rather than prospective protection evidence.

S39 Audited CFTC history-length extension

This section imports saved results from the separately developed final-jof branch. The local design commit precedes calculation; public upload occurred after calculation. This is not an externally registered or prospective financial confirmation.

The shorter-history outcomes had already been inspected. Only the longest-history forecasts were added under the fixed extension design. The same downloaded CFTC source arrays and reporting-vintage limitations apply.

Both low-history disadvantage screens pass. The longer-history advantage screen and strict sequential-improvement screen fail. The full regression has no registered high-region cell. The retained overall judgment is partial support.

Complete counterexample inventory. The following rows retain every saved counterexample. A market can appear under several screens. Loss changes are relative to the matched historical reference; their signs do not redefine a screen after observing it.

Table S52: Fixed history-length decision screens, including failed and untestable predictions. Counts use the original planned denominators.

Screen	Regression	Successes / planned	Required fraction	Judgment
P1	Four-feature	15 / 17	80%	Supported
P1	Full	29 / 34	80%	Supported
P2	Four-feature	10 / 17	60%	Not supported
P2	Full	—	—	Not testable
P3	Four-feature	6 / 17	70%	Not supported

Table S53: All saved counterexamples to the transported history screens.

Screen	Market code	History	Relative loss change (percent)
P1_QR4	240743	52	-7.324391
P1_QR4	098662	52	-0.437432
P2_QR4	044601	520	+0.811445
P2_QR4	043602	520	+8.026475
P2_QR4	042601	520	+1.137518
P2_QR4	045601	520	+1.520938
P2_QR4	099741	520	+0.618103
P2_QR4	095741	520	+3.875925
P2_QR4	232741	520	+0.575901
P1_QR33	045601	52	-2.041454
P1_QR33	1170E1	260	-16.215517
P1_QR33	090741	260	-3.599484
P1_QR33	33874A	260	-4.775476
P1_QR33	240743	260	-2.283380
P3_QR4	042601	52/260/520	+9.475655, +74.230130, +1.137518
P3_QR4	045601	52/260/520	+0.311844, +1.845046, +1.520938
P3_QR4	099741	52/260/520	+13.973619, +0.217492, +0.618103
P3_QR4	1170E1	52/260/520	+6.823794, -28.436244, -25.194932
P3_QR4	097741	52/260/520	+23.757442, -6.540749, -4.288684
P3_QR4	096742	52/260/520	+3.411816, +4.815826, -17.900542
P3_QR4	095741	52/260/520	+19.545430, +2.130022, +3.875925
P3_QR4	090741	52/260/520	+10.280676, -5.346398, -0.963088
P3_QR4	232741	52/260/520	+123.325833, -3.354286, +0.575901
P3_QR4	240743	52/260/520	-7.324391, -2.283380, -7.696425
P3_QR4	098662	52/260/520	-0.437432, -39.440706, -0.546595

Provenance and interpretation. The design was committed locally before the recorded execution start; the audit records both original commit identities and their UTC times. The saved summary retains its execution-stage verification marker; later acceptance and scalar-check receipts supply the completed audit status. Scalar agreement by the same executor is not independent dealer replication or another solver fit.

Full cells, candidate reports, execution receipts and source identities are preserved in the audited result archive. The import audit distinguishes original data from additional non-data code dependencies. These related-market counts establish neither a sufficient financial history nor a causal mechanism.

References

[1] Arjani N, Li F, Sabetti L. Monitoring intraday liquidity risks in a real time gross settlement system. Payments Canada; 2020. https://www.payments.ca/sites/default/files/discussion_paper_monitoring_intraday_liquidity_risks_in_a_real_time_gross_settlement_system.pdf.

[2] Barone-Adesi G, Giannopoulos K, Vosper L. VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets.* 1999;19(5):583–602. doi: 10.1002/(SICI)1096-9934(199908)19:5<583::AID-FUT5>3.0.CO;2-S

[3] Board of Governors of the Federal Reserve System. Reporting guidelines for preparing the FR 2004 primary government securities dealers reports. Board of Governors of the Federal Reserve System; 2013. https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=f555fe11-e149-479b-8088-4ed4e9ae1206.

[4] Board of Governors of the Federal Reserve System. Instructions for the preparation of government securities dealers reports: Reporting form FR 2004. Board of Governors of the Federal Reserve System; 2024. https://www.federalreserve.gov/apps/reportingforms/Download/DownloadAttachment?guid=950ab256-b485-4586-99b0-b27eb205fdfa.

[5] Boudoukh J, Richardson M, Stanton R, Whitelaw RF. MaxVaR: Long-horizon value at risk in a mark-to-market environment. *Journal of Investment Management.* 2004;2(3):1–6.

[6] Christoffersen PF. Evaluating interval forecasts. *International Economic Review.* 1998;39(4):841–862. doi: 10.2307/2527341

[7] Diebold FX, Mariano RS. Comparing predictive accuracy. *Journal of Business & Economic Statistics.* 1995;13(3):253–263. doi: 10.1080/07350015.1995.10524599

[8] Duffie D, Fleming MJ, Keane FM, Nelson C, Shachar O, Van Tassel P. Dealer capacity and U.S. Treasury market functionality. Federal Reserve Bank of New York Staff Reports; 2023. August 2023, revised October 2023. https://www.newyorkfed.org/research/staff_reports/sr1070

[9] Ehm W, Gneiting T, Jordan A, Krüger F. Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society Series B: Statistical Methodology.* 2016;78(3):505–562. doi: 10.1111/rssb.12154

[10] Engle RF, Manganelli S. CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics.* 2004;22(4):367–381. doi: 10.1198/073500104000000370

[11] Federal Reserve Bank of New York. Primary dealer statistics. Federal Reserve Bank of New York; n.d. https://www.newyorkfed.org/markets/counterparties/primary-dealers-statistics.

[12] Giacomini R, White H. Tests of conditional predictive ability. *Econometrica.* 2006;74(6):1545–1578. doi: 10.1111/j.1468-0262.2006.00718.x

[13] Gibbs I, Candès EJ. Adaptive conformal inference under distribution shift. In: Ranzato M, Beygelzimer A, Dauphin Y, Liang PS, Wortman Vaughan J, eds. *Advances in Neural Information Processing Systems*. 34. Curran Associates, Inc. 2021:1660–1672. https://proceedings.neurips.cc/paper_files/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html.

[14] Gneiting T. Making and evaluating point forecasts. *Journal of the American Statistical Association.* 2011;106(494):746–762. doi: 10.1198/jasa.2011.r10138

[15] Harvey D, Leybourne S, Newbold P. Testing the equality of prediction mean squared errors. *International Journal of Forecasting.* 1997;13(2):281–291. doi: 10.1016/s0169-2070(96)00719-4

[16] Holm S. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics.* 1979;6(2):65–70.

[17] Hull J, White A. Incorporating volatility updating into the historical simulation method for value-at-risk. *The Journal of Risk.* 1998;1(1):5–19. doi: 10.21314/jor.1998.001

[18] Koenker R, Bassett G. Regression quantiles. *Econometrica.* 1978;46(1):33–50. doi: 10.2307/1913643

[19] Kupiec PH. Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives.* 1995;3(2):73–84. doi: 10.3905/jod.1995.407942

[20] Lindsay D. Net position time series dataset. Kaggle; 2020. `https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset`.

[21] McNeil AJ, Frey R. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance.* 2000;7(3–4):271–300. doi: 10.1016/s0927-5398(00)00012-8

[22] Meinshausen N. Quantile regression forests. *Journal of Machine Learning Research.* 2006;7(35):983–999.

[23] Newey WK, West KD. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica.* 1987;55(3):703–708. doi: 10.2307/1913610

[24] Petrunia R, Sabetti L, Voia M. Tail risk in the automated clearing settlement system (ACSS). Payments Canada; 2019. `https://www.payments.ca/sites/default/files/tail_risk_in_the_automated_clearing_settlement_system_acss.pdf`.

[25] van den Hoogen S. Intraday liquidity risk estimation using transaction data: An extreme value theory approach. master's thesis. Delft University of Technology; 2017. `https://resolver.tudelft.nl/uuid:1c0c4545-bb07-4916-aa37-1d4bea3b7872`.

[26] Wisniewski W, Lindsay D, Lindsay S. Application of conformal prediction interval estimations to market makers' net positions. In: Gammerman A, Vovk V, Luo Z, Smirnov E, Cherubin G, eds. *Proceedings of the Ninth Symposium on Conformal and Probabilistic Prediction and Applications*. 128. PMLR 2020:285–301. `https://proceedings.mlr.press/v128/wisniewski20a.html`.

[27] Wisniewski WJ. *Net position and retail trader behaviour: Implications for market makers' risk management strategies*. Doctoral dissertation. Royal Holloway, University of London; 2024. `https://pure.royalholloway.ac.uk/en/publications/net-position-and-retail-trader-behaviour-implications-for-market-/`.

[28] Bollerslev T. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics.* 1986;31(3):307–327. doi: 10.1016/0304-4076(86)90063-1

[29] Hansen BE. Testing for parameter instability in linear models. *Journal of Policy Modeling.* 1992;14(4):517–533. doi: 10.1016/0161-8938(92)90019-9

[30] Hastie T, Tibshirani R, Friedman J. *The elements of statistical learning: Data mining, inference, and prediction*. Springer. 2 ed., 2009

[31] Hansen PR, Lunde A, Nason JM. The Model Confidence Set. *Econometrica*. 2011;79(2):453–497. doi: 10.3982/ECTA5771

[32] Avellaneda M, Stoikov S. High-frequency trading in a limit order book. *Quantitative Finance*. 2008;8(3):217–224. doi: 10.1080/14697680701381228

[33] Commodity Futures Trading Commission. Historical compressed. n.d. Retrieved October 24, 2026. `https://www.cftc.gov/MarketReports/CommitmentsofTraders/HistoricalCompressed/index.htm`.